Bright Tuition classs

Subject : Biology, Physics, Chemistry
Standard : 11
Total Mark : 720

NEET practice 15 th march

Paper Set : 1
Date : 15-03-2026
Time : 3H:0M

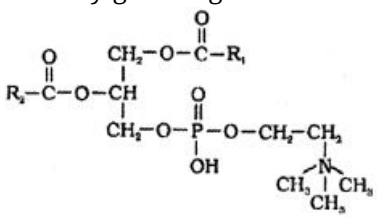
Biology - Section A (MCQ)

(1) Black rust of wheat is caused by
(A) Puccinia graminis (B) Ustilago
(C) Pythium (D) None of these

(2) A : Chitinous exoskeleton is a characteristic feature of arthropods.

R : It allows diffusion of water vapour from atmosphere to the body.
(A) Assertion and Reason both are correct and also correct explanation.
(B) Assertion and Reason both are correct but not explanation of assertion.
(C) Assertion is correct, but Reason is incorrect.
(D) Both Assertion and Reason are incorrect.

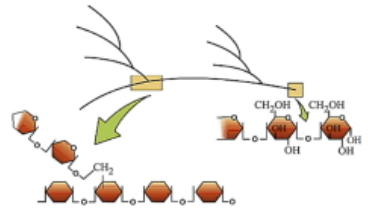
(3) Identify given figure.



(A) Cholesterol (B) Triglycerides
(C) Lecithin (D) Adinylic acid

(4) Which of the following respires by gills
(A) Prawn (B) Frog
(C) Crocodile (D) Whale

(5) Which one is not correct for given structure of glycogen.



(A) There is only one reducing end
(B) Total six a$1-6$ glycosidic bond are present
(C) $\alpha 1-6$ glycosidic bond is more than $\alpha 1-4$ glycosidic bond
(D) each branch end with Non reducing end.

(6) Choose correct pair.

$Column-I$	$Column-II$
1.$Phaeophyceae$	$A.Chara$
2.$Chlorophyceae$	$B.Wolffia$
3.$Rhodophyceae$	$C.Porphyra$
4.$Angiosperms$	$D.Ectocarpus$

(A) $(1-D),(2-A),(3-B),(4-C)$
(B) $(1-D),(2-C),(3-B),(4-A)$
(C) $(1-D),(2-A),(3-C),(4-B)$
(D) $(1-C),(2-A),(3-D),(4-B)$

(7) Which of the following $statement(s)$ is/are correct for Metabolism?
(A) It is the sum total of all physical reactions taking place inside a living system.
(B) All plants, animals, fungi and microbes exhibit metabolism.
(C) Isolated metabolic reactions in-vitro are not living but are living reactions.
(D) All of the above

(8) Read the following statement and choose the correct characteristic feature of Aves
(A) Skin of birds have glands
(B) Air sacs help in excretion
(C) Heart is having three auricle and one ventricle
(D) Preen gland is present at the base of tail

(9) Growth in living organisms occurs by
(A) division of cells.
(B) increase in biomass.
(C) accumulation of materrial by external agency.
(D) both (a) and (b).

(10) Which of the following mode of respiration always continue in frog?
(A) buccopharyngeal (B) Pulmonary
(C) Cutaneous (D) Gills

(11) Sexual reproduction in algae results in the formation of
(A) Oospore (B) Zoospore
(C) Zygote (D) Zygospore

(12) Microbodies differ from lysosomes in that

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(A) Microbodies are surrounded by a single unit membrane while lysosome membrane is double

(B) Microbodies are surrounded by double membrane while lysosomes membrane is single unit

(C) Microbodies contain lytic enzymes while lysosomes do not

(D) Lysosome contain lytic enzymes while microbodies do not

(13) Bryophytes resemble algae in the following aspect.

(A) Filamentous body, presence of vascular tissues and autotrophic nutrition

(B) Differentiation of plant body into root, stem and leaves and autotrophic nutrition

(C) Thallus like plant body, presence of roots and autotrophic nutrition

(D) Thallus like plant body, lack of vascular tissues and autotrophic nutrition

(14) A nucleotide has three chemically distinct compounds. These are A, B and C

(A) A - Sugar, B-carbonates, C-chlorides

(B) A-DNA, B-cellulose, C-chitin

(C) A-Purine or pyrimidine B-Monosaccharide, C-Phosphate

(D) A-Phosphoric acid, B-Proteins, C-acids

(15) It is not part of flower.

(A) Gynoecium (B) Androecium

(C) Bract (D) Corolla

(16) Branchiostoma is an example of

(A) Cyclostomata (B) Chondrichthyes

(C) Cephalochordata (D) Osteichthyes

(17) Vegetative reproduction by fragmentation is common in

(A) Agaricus (B) Saccharomyces

(C) Euglena (D) Gonyaulax

(18) The cell wall is impermeable to water and deposition of suberin is also found in

(A) Bast (B) Cork

(C) Bark (D) Xylem

(19) Choose correct pair.

(A) Nitrogen base - alanine

(B) Polysaccharide - glycogen

(C) Amino acid - trypsin

(D) Enzyme - insulin

(20) Identify the plant parts whose transverse section shows a clear and prominent pith.

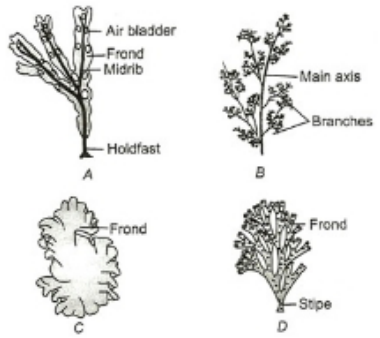
(A) Dicot and monocot stems

(B) Dicot stem and monocot root

(C) Dicot and monocot roots

(D) Dicot stem and dicot root

(21) Identify the given figures of algae and select the correct option



(A) $A - Volvox, B - Chlamydomonas, C - Chara, D - Porphyra$

(B) $A - Fucus, B - Polysiphonia, C - Porphyra, D - Dictyota$

(C) $A - Fucus, B - Dictyota, C - Porphyra, D - Polysiphonia$

(D) $A - Dictyota, B - Porphyra, C - Fucus, D - Polysiphonia$

(22) Match the following column I with coloumn II and choose the correct combination.

Column $-I$	Column $-II$
(a) Earthworm	(i) Gizzard
(b) Cockroach	(ii) Caecum
(c) Frog	(iii) Clitellum
(d) Rat	(iv) Cloaca

(A) $(a) - (i)$, $(b) - (ii)$, $(c) - (iv)$, $(d) - (iii)$

(B) $(a) - (iii)$, $(b) - (i)$, $(c) - (iv)$, $(d) - (ii)$

(C) $(a) - (iii)$, $(b) - (i)$, $(c) - (ii)$, $(d) - (iv)$

(D) $(a) - (i)$, $(b) - (iii)$, $(c) - (iv)$, $(d) - (ii)$

(23) When fungi feed on dead organic matter, they are known as

(A) Dimorphic (B) Parasites

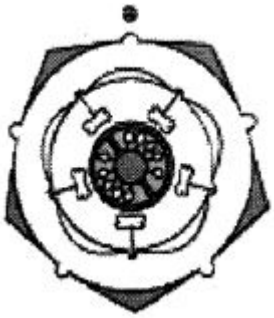
(C) Saprophytes (D) None of these

(24) Select the correct floral diagram of Brassicaceae

(A)

(B)

(C)



(D)

(25) In which the pneumatophores are found

(A) Tinospora (B) Pinus

(C) Rhizophora (D) None of these

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(26) Stems are modified into flattened structures, which carryout photosynthesis in

(A) Euphorbia (B) Opuntia

(C) Bougainvillea (D) Colocasia

(27) Identify the correct option for Cyclostomata : –

(A) Development indirect, $6-15$ pairs of gillslits, ectoparasites on some fishes

(B) Development indirect, viviparous, $6-15$ pairs of gill–slits, scales and paired fins present.

(C) Jawless, scales absent, only paired fins are present

(D) Both paired and unpaired fins are present, ventral mouth, $6-15$ pairs of gill–slits

(28) Which of the following is not a significance of scientific nomenclature?

(A) It ensure that one organism is recognised by one scientific name only.

(B) One scientific name is given to one organism only.

(C) Description of one organism leads to two scientific name all around the world.

(D) 1 and 2 both

(29) Systematics refers to the

(A) identification and classification of plants and animals.

(B) nomenclature and identification of plants and animals.

(C) diversity of kinds of organisms and their relationship.

(D) different kinds of organisms and their classification.

(30) Aerenchyma is helpful in plants by

(A) Giving flexibility to plants

(B) Giving mechanical strength to plants

(C) Promoting photosynthesis

(D) Providing buoyancy in hydrophytes

(31) Ornithorhynchus is a

(A) A fossil bird (B) Duck

(C) Monotreme mammal (D) Dinosaur

(32) I. Made up of sclerenchymatous cells

II. Generally absent in primary phloem

III. Much elongated and pointed

Given above characters belongs to which of the following?

(A) Phloem fibre (B) Xylem fibre

(C) Companion cells (D) Sieve cells

(33) The roots of gymnosperms is

(A) Tap root (B) Adventitious roots

(C) Rhizoid (D) Rhizophores

(34) A : The inner lining of trachea, bronchi and fallopian tubes has simple epithelium with non-motile cilia.

R : Cilia form the brush-border.

(A) Assertion and Reason both are correct and also correct explanation.

(B) Assertion and Reason both are correct but not explanation of assertion.

(C) Assertion is correct, but Reason is incorrect.

(D) Both Assertion and Reason are incorrect.

(35) Ribosome mainly has

(A) DNA (B) RNA

(C) Carbohydrate (D) None of these

(36) Zygospores are formed in

(A) Puccinia (B) Penicillium

(C) Alternaria (D) Mucor / Rhizopus

(37) The practical purpose of classification of living organisms is to

(A) explain the origin of living organsims.

(B) trace the evolution of living organsims.

(C) name the living organisms.

(D) facilitate identification of unknown organisms.

(38) Which of the following pairs come under the group chrysophytes?

(A) Diatoms and Euglena

(B) Euglena and Trypanosoma

(C) Diatoms and Desmids

(D) Gonyaulax and Desmids

(39) Roots are used in vegetative propagation in

(A) Potato (B) Sweet potato

(C) Ginger (D) Onion

(40) Largest phylum, jointed appendages, chitinous exoskeleton, mostly dioecious The above features belongs to : –

(A) Pheretima (B) Fasciola

(C) Apis (D) Ascaris

(41) Which is not an example of prokaryotic cell ?

(A) Bacteria (B) Mycoplasma

(C) Egg of ostrich (D) $PPLO$

(42) In Gracilaria, sexual reproduction is

(A) Isogamous (B) Anisogamous

(C) Oogamous (D) Both (a) and (b)

(43) Given organisms belongs to how many genera?

Wheat, Brinjal, Potato, Lion, Dog, Tiger

(A) Three (B) Two

(C) Four (D) Five

(44) Epipetalous androecium is found in

(A) Mung (B) Pea

(C) Solanum nigrum (D) Groundnut

(45) Stomata are the component of

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(A) Epidermal tissue system

(B) Ground tissue system

(C) Conducting tissue system

(D) Vascular tissue system

(46) The structure of nuclear membrane facilitates

(A) Synapsis of homologous chromosomes at meiosis

(B) Nucleo-cytoplasmic exchange of materials

(C) Anaphasic separation of daughter chromosomes

(D) Organization of spindles

(47) DNA is

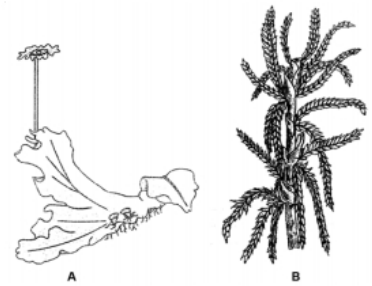
(A) Always double stranded

(B) Rarely single stranded

(C) Always single stranded

(D) Rarely double stranded

(48) Identify the plants A and B in the figures given below



(A) A - Female Marchantia, B - Sphagnum

(B) A - Female Marchantia, B - Sphagnum

(C) A - Marchantia, B - Funaria

(D) A - Male Marchantia, B - Sphagnum

(49) Which of the following is incorrect?

(A) Mycoplasma is the smallest cell

(B) Bacteria are 3 to 5 μm in size.

(C) The largest cell is the egg of an ostrich.

(D) Nerve cells are some of the smallest cells.

(50) Dominant protein in biosphere is

(A) Trypsin (B) Rubisco

(C) Collagen (D) Keratin

(51) Match the following:

Column - I	Column - II
1. Smooth muscles	p. Hard and not pliable ground substance
2. Skeletal muscles	q. Blood vessel stomach intestine
3. Bone	r. Condroblast
4. Cartilage	s. A-band

(A) $(1-p),(2-q),(3-r),(4-s)$

(B) $(1-q),(2-p),(3-s),(4-r)$

(C) $(1-q),(2-s),(3-p),(4-r)$

(D) $(1-p),(2-r),(3-s),(4-q)$

(52) Vessels are mainly found in

(A) Xylem of angiosperms (B) Xylem of gymnosperms

(C) Both (a) and (b) (D) None of these

(53) The calyptrogen of the root apex forms

(A) Rhizoids (B) Root nodule

(C) Root hairs (D) Root cap

(54) Mark the odd one is each series :

(a) Fat body, Nephrocyte, Urinary bladder (in cockroach)

(b) Blood, Bone, Cartilage, Muscle

(c) Skeletal, Cardiac, Nerve, Smooth Muscles

(A) $(a)-$Nephrocyte, $(b)-$Muscle, $(c)-$Nerve

(B) $(a)-$Fat body, $(b)-$Cartilage, $(c)-$Cardiac Muscle

(C) $(a)-$Nephrocyte, $(b)-$Bone, $(c)-$Skeletal Muscle

(D) $(a)-$Urinary Bladder, $(b)-$Muscle, $(c)-$Nerve

(55) Chlorophyll containing fleshy cylindrical structures found in Euphorbia are modified

(A) Roots (B) Fruit

(C) Leaves (D) Stem

(56) An example of agarophyte is

(A) Dictyota (B) Fucus

(C) Nostoc (D) Gelidium

(57) Which covers the largest number of organisms

(A) Genus (B) Family

(C) Phylum (D) Class

(58) Archaebacteria found in extremely salty area

(A) Halophiles (B) Methanogens

(C) Thermoacidophiles (D) Cyanobacteria

(59) Starch and cellulose are the compounds made up of many units of

(A) Simple sugar (B) Fatty acid

(C) Glycerol (D) Amino acid

(60) Choose the correct names of the different bacteria given below according to their shapes.

(A) A- Cocci, B - Bacilli, C - Spirilla, D - Vibrio

(B) A - Bacilli, B- Cocci, C - Spirilla, D - Vibrio

(C) $A-$ Spirilla, B - Bacilli, C - Cocci, D - Vibrio

(D) $A-$ Spirilla, B - Vibrio, C- Cocci, D - Bacilli

(61) Non poisonous snake is : $-$

(A) Vipera (B) Naja

(C) Bangarus (D) Azgar

(62) How many statements are correct?

(A) Animal cells contain membrane bound organelle called centriole which help in cell division.

(B) Lipids are arranged within the plasma membrane with the polar head towards the inner sides and the hydrophobic tails towards the outer side.

(C) The cell wall and middle lamellae may be traversed by plasmodesmata which connect the cytoplasm of neighboring plant cells.

(D) The quasifluid nature of lipid enables lateral movement of proteins within the overall bilayer.

(A) 1 (B) 2

(C) 3 (D) 4

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(63) What is correct for schlerenchyma tissue

(A) Intercellular space, oval shaped cell, photosynthesis, storage and secretion are functions of its

(B) Cells may be oval, spherical or polygonal and contain chlorophyll, living cells.

(C) Dead cells and without protoplast, provides mechanical support

(D) Cell wall is thin and made up of cellulose, intercellular space is present

(64) Fishes, amphibians, reptiles and birds are kept in the same

(A) Order (B) Class

(C) Genus (D) Phylum

(65) Stilt roots are reported from

(A) Pandanus (B) Radish

(C) Mango-ginger (D) Bryophyllum

(66) Identify the correct statement regarding enzymes

(A) Enzymes are specific biological catalysts that can normally function at very high temperatures ($T1000\ K$)

(B) Enzymes are normally heterogeneous catalysts that are very specific in their action

(C) Enzymes are specific biological catalysts that cannot be poisoned

(D) Enzymes are specific biological catalysts that possess well defined active sites

(67) Frog which lives on the trees

(A) Alytes (B) Bufo

(C) Hyla (D) Rana

(68) Intercalary meristem results in

(A) Secondary growth (B) Primary growth

(C) Apical growth (D) Periderm formation

(69) Scientific name of wheat.

(A) Triticum aestivum (B) Anacardiaceae

(C) Musca domestica (D) Mangifera indica

(70) What are the three basic components of connective tissues?

(A) Ground substance, cells and basement membrane

(B) Cartilage, intercellular matrix and serum

(C) Cells, protein fibers and ground substance

(D) Collagen, elastin and reticular fibers

(71) Which of the following is correct for middle lamella of eukaryotic cell ?

(A) It is formed in plant cell

(B) it is mainly consists of $Ca-$ pectate

(C) It holds different neighbouring cells together

(D) All of the above

(72) Blind sac body plan is shown by

(A) Roundworms (B) Annelids

(C) Coelenterates (D) Arthropods

(73) Aestivation in cassia calotropis is respectively.....

(A) Imbricate, Valvate (B) Vexillary, twisted

(C) Valvate, Twisted (D) Vexillary, valvate

(74) Identify the components labelled as A, B, C, D and E in the diagram given below from the list (i) to $(viii)$.

(i) Cristae of mitochondria (ii) Inner membrane of mitochondria (iii) Cytoplasm (iv) Smooth endoplasmic reticulum (v) Rough endoplasmic reticulum (vi) Mitochondrial matrix (vii) Ribosome $(viii)$ Nucleus

(A) $(viii)$ (v) (vii) (iii) (iv) (B) (i) (iv) (vii) (vi) (iii)

(C) (vi) (v) (iv) (vii) (i) (D) (v) (i) (iii) (ii) (iv)

(75) Mast cells are found in

(A) Connective tissue (B) Muscular tissue

(C) Nervous tissue (D) Blood

(76) Lacunae are connected with

(A) Canaliculae (B) Sublacunae

(C) Both (a) and (b) (D) None of the above

(77) No cell could live without

(A) Phytochrome (B) Enzymes

(C) Chloroplasts (D) Protein

(78) Which components present in plasma membrane as result it having fluid structure?

(A) Carbohydrates (B) Lipid

(C) Glycoprotein (D) Polysaccherricles

(79) Statement$-I$: Nuclear membrane is absent in bacteria.

Statement$-II$: Methanogens are responsible for the production of methane.

(A) Both statements I and II are correct

(B) Only statement I is incorrect

(C) Both statements I and II are incorrect

(D) Only statement II is incorrect

(80)

Column-I (Terms)	Column-ll (Examples)
A. Ascus	I. Spirulina
B. Basidium	II. Penicillium
C. Protista	III. Agaricus
D. Cyanobacteria	IV. Euglena
E. Animalia	V. Sponges

Match the terms given in column-I with their examples given in column-II and choose the correct option

(A) $A-II, B-III, C-IV, D-V, E-I$

(B) $A-I, B-II, C-III, D-V, E-IV$

(C) $A-II, B-V, C-III, D-I, E-IV$

(D) $A-II, B-III, C-IV, D-I, E-V$

(81) In Cockroach, the number of ganglia are

(A) Two pairs thoracic and four pairs abdominal

(B) Three pairs thoracic and six pairs abdominal

(C) Three pairs thoracic and five pairs abdominal

(D) Two pairs thoracic and six pairs abdominal

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(82) Coralloid roots have a symbiotic association with

(A) Photosynthetic green algae

(B) Nitrogen-fixing cyanobacteria

(C) Fungus

(D) Photosynthetic brown algae

(83) Which of the following statements regarding cyanobacteria is incorrect?

(A) It is also called blue green algae.

(B) They are chemosynthetic autotrophs.

(C) It forms blooms in polluted water bodies.

(D) It is unicellular, colonial or filamentous, marine or terrestrial bacteria.

(84) Meristem is defined as a plant tissue where

(A) Cell conserves food and supply it to new ones

(B) Cells mature and add to the bulk of a plant

(C) Cells elongate and add to the growth of a plant

(D) Cells divide continuously to give rise to new ones

(85) Which of the following stage of Funaria is haploid

(A) Gametophyte (B) Sporophyte

(C) Both (a) and (b) (D) None of these

(86) Layer of cells between endodermis and vascular bundles is called

(A) Epidermis (B) Pericycle

(C) Hypodermis (D) Pith

(87) Frog can protect itself from enemies by ?

(A) Teeth (B) Camouflage

(C) Clawed digit (D) Both (1) and (2)

(88) When centromere present slightly away from the middle of the chromosome, then chromosome is called : –

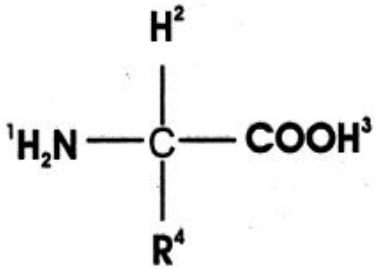
(A) Metacentric chromosome

(B) Sub–metacentric chromosome

(C) Acrocentric chromosome

(D) Telocentric chromosome

(89) Which of the two groups of following formula is involved in peptide bond formation between different amino acids?



(A) 1 and 3 (B) 2 and 4

(C) 2 and 3 (D) 1 and 4

(90) ..... is family of cat.

(A) Felidae (B) Canidae

(C) Hominidae (D) Muscidae

Physics - Section A (MCQ)

(91) Two forces $F_1 = 1\,N$ and $F_2 = 2\,N$ act along the lines $x = 0$ and $y = 0$ respectively. Then the resultant of forces would be

(A) $\hat{i} + 2\hat{j}$ (B) $\hat{i} + \hat{j}$

(C) $3\hat{i} + 2\hat{j}$ (D) $2\hat{i} + \hat{j}$

(92) Two forces $\vec{F}_1 = 5\hat{i} + 10\hat{j} - 20\hat{k}$ and $\vec{F}_2 = 10\hat{i} - 5\hat{j} - 15\hat{k}$ act on a single point. The angle between $\vec{F}_1$ and $\vec{F}_2$ is nearly _____ $^\circ$.

(A) 30 (B) 45

(C) 60 (D) 90

(93) A body A starts from rest with an acceleration a_1 . After $2\ seconds$, another body B starts from rest with an acceleration a_2 . If they travel equal distance in the $5th\ second$, after the start of A , then the ratio a_1 : a_2 is equal to

(A) $5:9$ (B) $5:7$

(C) $9:5$ (D) $9:7$

(94) Wind is blowing in the north direction at speed of $2\ m/s$ which causes the rain to fall at some angle with the vertical. With what velocity should a cyclist drive so that the rain appears vertical to him :

(A) $2\ m/s$ south (B) $2\ m/s$ north

(C) $4\ m/s$ west (D) $4\ m/s$ south

(95) Two forces of $10\,N$ and $6\,N$ act upon a body. The direction of the forces are unknown. The resultant force on the body may be .....N

(A) 15 (B) 3

(C) 17 (D) 2

(96) A body is moving in a circular path with a constant speed. It has

(A) A constant velocity

(B) A constant acceleration

(C) An acceleration of constant magnitude

(D) An acceleration which varies with time

(97) Given $\left|\vec{A}_1\right| = 2,\ \left|\vec{A}_2\right| = 3$ and $\left|\vec{A}_1 + \vec{A}_2\right| = 3$. Find the value or $\left|\left(\vec{A}_1 + 2\vec{A}_2\right) \times \left(3\vec{A}_1 - 4\vec{A}_2\right)\right|$

(A) 64 (B) 60

(C) 62 (D) 61

(98) Which of the following is a scalar quantity

(A) Displacement (B) Electric field

(C) Acceleration (D) Work

(99) A man projects a coin upwards from the gate of a uniformly moving train. The path of coin for the man will be

(A) Parabolic (B) Inclined straight line

(C) Vertical straight line (D) Horizontal straight line

(100) If the length of the second's hand in a stop clock is $3\,cm$ the angular velocity and linear velocity of the tip is

(A) $0.2047\,rad/sec, 0.0314\,m/sec$

(B) $0.2547\,rad/sec, 0.314\,m/sec$

(C) $0.1472\,rad/sec, 0.06314\,m/sec$

(D) $0.1047\,rad/sec, 0.00314\,m/sec$

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(101) Let $\vec{v}$ and $\vec{a}$ denote the velocity and acceleration respectively of a body in one-dimensional motion

(A) $|\vec{v}|$must decrease when $\vec{a} < 0$

(B) Speed will decrease when $|\vec{v}| < 0$ and $\vec{a}$ are < 0

(C) Speed will increase when both $|\vec{v}|$ and $\vec{a}$ are < 0

(D) Both (B) and (C)

(102) For component of a vector $A = (3\hat{i} + 4\hat{j} - 5\hat{k})$, match the following colum.

Colum I	Colum II
(A) $x-$axis	(p) $5\,unit$
(B) Along another vector $(2\hat{i} + \hat{j} + 2\hat{k})$	(q) $4\,unit$
(C) Along $(6\hat{i} + 8\hat{j} - 10\hat{k})$	(r) 0
(D) Along another vector $(-3\hat{i} - 4\hat{j} + 5\hat{k})$	(s) None

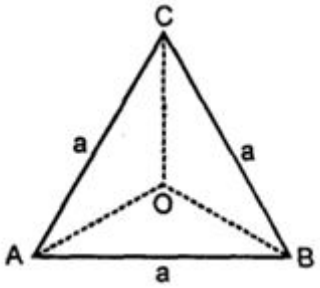
(A) $(A \to q, B \to r, C \to s, D \to s)$

(B) $(A \to p, B \to r, C \to s, D \to s)$

(C) $(A \to r, B \to q, C \to s, D \to s)$

(D) $(A \to q, B \to r, C \to s, D \to p)$

(103) ABC is an equilateral triangle. Length of each side is a and centroid is point O. Find $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = ...$



(A) 0

(B) 1

(C) 2

(D) 3

(104) A cricketer can throw a ball to a maximum horizontal distance of $100\,m$. With the same effort, he throws the ball vertically upwards. The maximum height attained by the ball is ..... m

(A) 100

(B) 80

(C) 60

(D) 50

(105) A man moves in an open field such that after moving $10\,m$ on a straight line, he makes a sharp turn of $60°$ to his left. The total displacement just at the start of 8th turn is equal to .....m

(A) 12

(B) 15

(C) 17.32

(D) 14.14

(106) Two forces, F_1 and F_2 are acting on a body. One force is double that of the other force and the resultant is equal to the greater force. Then the angle between the two forces is

(A) $\cos^{-1}(1/2)$

(B) $\cos^{-1}(-1/2)$

(C) $\cos^{-1}(-1/4)$

(D) $\cos^{-1}(1/4)$

(107) The displacement of a particle is proportional to the cube of time elapsed. How does the acceleration of the particle depends on time obtained

(A) $a \propto t^2$

(B) $a \propto 2t$

(C) $a \propto t^3$

(D) $a \propto t$

(108) The magnitude of the X and Y component of $\vec{A}$ are 7 and 6 respectively. Also the magnitude of X and Y component of $\vec{A} + \vec{B}$ are 11 and 9 respectively. What is the magnitude of $\vec{B}$?

(A) 5

(B) 6

(C) 8

(D) 9

(109) A particle revolves round a circular path. The acceleration of the particle is

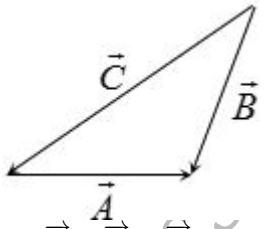
(A) Along the circumference of the circle

(B) Along the tangent

(C) Along the radius

(D) Zero

(110) For the figure



(A) $\vec{A} + \vec{B} = \vec{C}$

(B) $\vec{B} + \vec{C} = \vec{A}$

(C) $\vec{C} + \vec{A} = \vec{B}$

(D) $\vec{A} + \vec{B} + \vec{C} = 0$

(111) A ball dropped from the top of tower falls first half height of tower in $10\,s$. The total time spend by ball in air is ..... s [Take $g = 10\,m/s^2$]

(A) 14.14

(B) 15.25

(C) 12.36

(D) 17.36

(112) A ball of mass m is thrown vertically upwards.Assume the force of air resistance has magnitude proportional to the velocity, and direction opposite to the velocity's. At the highest point, the ball's acceleration is

(A) 0

(B) less than g

(C) g

(D) greater than g

(113) The value of a unit vector in the direction of vector $A = 5\hat{i} - 12\hat{j}$, is

(A) $\hat{i}$

(B) $\hat{j}$

(C) $(\hat{i} + \hat{j})/13$

(D) $(5\hat{i} - 12\hat{j})/13$

(114) A particle initially at rest is subjected to two forces. One is constant, the other is a retarding force proportional to the particle velocity. In the subsequent motion of the particle

(A) the acceleration will increase from zero to a constant value

(B) the acceleration will decrease from its initial value to zero

(C) the velocity will increase from zero to a constant value.

(D) Both (B) and (C)

(115) A ball is thrown upwards from the top of a tower $40\,m$ high with a velocity of $10\,m/s$, find the time when it strikes the ground .....sec : $(g = 10\,m/s^2)$

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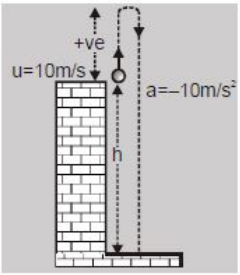


(A) 8 (B) 1

(C) 4 (D) 6

(116) A projectile is thrown upward with a velocity v_0 at an angle α to the horizontal. The change in velocity of the projectile when it strikes the same horizontal plane is

(A) $v_0 \sin\alpha$ vertically downwards

(B) $2v_0 \sin\alpha$ vertically downwards

(C) $2v_0 \sin\alpha$ vertically upwards

(D) zero

(117) Given that; $A = B = C$. If $\vec{A} + \vec{B} = \vec{C}$, then the angle between $\vec{A}$ and $\vec{C}$ is θ_1. If $\vec{A} + \vec{B} + \vec{C} = 0$, then the angle between $\vec{A}$ and $\vec{C}$ is θ_2. What is the relation between θ_1 and θ_2 ?

(A) $\theta_1 = \theta_2$ (B) $\theta_1 = \theta_2/2$

(C) $\theta_1 = 2\theta_2$ (D) None of these

(118) Two bodies of masses $1kg$ and $5kg$ are dropped gently from the top of a tower. At a point $20cm$ from the ground, both the bodies will have the same

(A) Momentum (B) Kinetic energy

(C) Velocity (D) Total energy

(119) A particle is rotating in a circle of radius $1\,m$ with constant speed $4\,m/s$. In time $1\,s$, match the following (in SI units) columns.

Colum I	Colum II
(A) Displacement	(p) $8\sin 2$
(B) Distance	(q) 4
(C) Average velocity	(r) $2\sin 2$
(D) Average acceleration	(s) $4\sin 2$

(A) $(A \to r, B \to q, C \to r, D \to p)$

(B) $(A \to p, B \to q, C \to r, D \to p)$

(C) $(A \to r, B \to s, C \to r, D \to p)$

(D) $(A \to p, B \to q, C \to r, D \to s)$

(120) A particle is moving with constant speed $\sqrt{2}\,m/s$ on a circular path of radius $10\,cm$. Find the magnitude of average velocity when it has covered $\left(\frac{3}{4}\right)^{th}$ circular path

(A) $\frac{\pi}{3}\,m/s$ (B) $\frac{3}{2\pi}\,m/s$

(C) $\frac{3}{\pi}\,m/s$ (D) $\frac{4}{3\pi}\,m/s$

(121) Mark the wrong statement :-

(A) Nothing is in the state of absolute rest or stateof absolute motion.

(B) Magnitude of displacement is always equal to the distance travelled.

(C) Magnitude of displacement can never be greater than the distance travelled.

(D) Magnitude of instantaneous velocity is equal to the instantaneous speed.

(122) A particle is moving in a circle with uniform speed v. In moving from a point to another diametrically opposite point

(A) The momentum changes by mv

(B) The momentum changes by $2mv$

(C) The kinetic energy changes by $(1/2)mv^2$

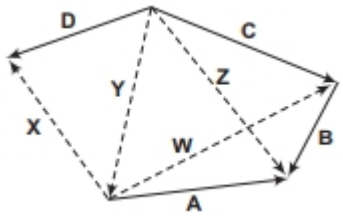
(D) The kinetic energy changes by mv^2

(123) A string of length $2\,m$ can withstand a maximum tension of $16000\,N$. A particle of mass $2\,kg$ is tied to one end of the string and rotated in a horizontal circle. What is the maximum number of rotations per second the particle can make so that the string does not break? $[\pi^2 = g]$

(A) 10 (B) 20

(C) 30 (D) 40

(124) In the diagram shown in figure



(A) $X = A + B - C + D$ (B) $Y = B + C - A$

(C) $Z = B + C$ (D) Only (b) and (c)

(125) The value of $\hat{i} \times (\hat{i} \times a) + \hat{j} \times (\hat{j} \times a) + \hat{k} \times (\hat{k} \times a)$ is

(A) a (B) $a \times \hat{k}$

(C) $-2\,a$ (D) $-a$

(126) Two bodies of mass $10\,kg$ and $5\,kg$ moving in concentric orbits of radii R and r such that their periods are the same. Then the ratio between their centripetal acceleration is

(A) R/r (B) r/R

(C) R^2/r^2 (D) r^2/R^2

(127) A car travels the first half of a distance between two places at a speed of $30\,km/hr$ and the second half of the distance at $50\,km/hr$. The average speed of the car for the whole journey is.....km/hr

(A) 42.5 (B) 40.0

(C) 37.5 (D) 35.0

(128) A ball is dropped downwards. After 1 second another ball is dropped downwards from the same point. What is the distance between them after $3\,seconds$.....m

(A) 25 (B) 20

(C) 50 (D) 9.8

(129) In the $s - t$ equation $\left(s = 10 + 20t - 5t^2\right)$, match the following columns.

Bright Tuition class

Colum I	Colum II
(A) Distance travelled in $3\,s$	(p) -20 units
(B) Displacement in $1\,s$	(q) 15 units
(C) Initial acceleration	(r) 25 units
(D) Velocity at $4\,s$	(s) -10 units

(A) $(A \to r, B \to p, C \to s, D \to p)$

(B) $(A \to p, B \to p, C \to s, D \to r)$

(C) $(A \to r, B \to s, C \to p, D \to p)$

(D) $(A \to s, B \to p, C \to r, D \to p)$

(130) A car accelerates from rest at a constant rate of $2\,m/s^2$ for some time. Then it retards at a constant rate $4\,m/s^2$ and comes to rest. If it remains in motion for 3 second then displacement covered by it ....m

(A) 4 (B) 6

(C) 8 (D) 10

(131) Two particles are at rest on a straight track and they are separated from each other by $100\,m$. They start their motion in the same direction, with the same acceleration $10\,m/s^{-2}$. Their relative displacement after $10\,s$ will be

(A) more than $100\,m$

(B) $100\,m$

(C) less than $100\,m$ but not zero

(D) zero

(132) A disc rotating about its axis from rest, acquires the angular speed $100\,rev/s$ in 4 second. The angle rotated by it during these four seconds (in radian) is ..... π

(A) 100 (B) 200

(C) 300 (D) 400

(133) The angle between two vectors given by $6\hat{i} + 6\hat{j} - 3\hat{k}$ and $7\hat{i} + 4\hat{j} + 4\hat{k}$ is

(A) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (B) $\cos^{-1}\left(\frac{5}{\sqrt{3}}\right)$

(C) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (D) $\sin^{-1}\left(\frac{\sqrt{5}}{3}\right)$

(134) The average acceleration vector for a particle having a uniform circular motion is

(A) A constant vector of magnitude $\frac{v^2}{r}$

(B) A vector of magnitude $\frac{v^2}{r}$ directed normal to the plane of the given uniform circular motion

(C) Equal to the instantaneous acceleration vector at the start of the motion

(D) A null vector

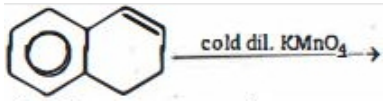
(135) The position of a particle moving along the $y-$ axis is given as $y = 3t^2 - t^3$ where y is in $metre$ and t is in $second$. The time when the particle attains maximum positive position will be.....s

(A) 1.5 (B) 4

(C) 2 (D) 3

Chemistry - Section A (MCQ)

(136) Product of the reaction is



(A) Meso compound (B) Enantiomeric pair

(C) Diastereomers (D) Optically pure enantiomer

(137) In a balanced equation $H_2SO_4 + x\,HI \to H_2S + y\,I_2 + z\,H_2O$, the values of x, y, z are

(A) $x = 3, y = 5, z = 2$ (B) $x = 4, y = 8, z = 5$

(C) $x = 8, y = 4, z = 4$ (D) $x = 5, y = 3, z = 4$

(138) Which of the following statement(s) is/are correct.
(A) The pH of $1 \times 10^{-8} M$ aqueous solution of HCl is 8
(B) The conjugate base of $H_2PO_4^{\ominus}$ is HPO_4^{2-}
(C) K_w increases with increase in temperature
(D) When a solution of weak monoprotic acid is titrated against a strong base then at half neutralisation point, $pH = \frac{1}{2}pK_a$.

(A) A, D (B) B, C

(C) A, B, C (D) B, C, D

(139) What will be the standard heat of reaction for the reaction for the reaction $C_2H_{6(g)} + 7/2O_{2(g)} \to 2CO_{2(g)} + 3H_2O_{(l)}$ if standard heat of combustion of $C_{(s)}$ and $H_{2(g)}$ are -393.5 and $-285\,kJ/mol$ respectively and standard heat of formation of $C_2H_{6(g)}$ is $-88.2\,kJ/mol$.....kJ/mol

(A) $+1553.8$ (B) -1730.2

(C) -1553.8 (D) -1456.6

(140) When 2-bromobutane reacts with alcoholic KOH, the reaction is called

(A) Halogenation (B) Hydrogenation

(C) Chlorination (D) Dehydro-halogenation

(141) $(A) CH_3 - CH_2 - CH_2 - CO_2H$ $(B) CH_3 - \underset{\underset{CO_2H}{|}}{CH} - CH_3$

(A) and (B) are

(A) chain isomer (B) position isomer

(C) functional isomer (D) metamers

(142) The lattice energy of $NaCl$ is $-780\,kJ\,mol^{-1}$. The enthalpies of hydration of $Na^+_{(g)}$ and $Cl^-_{(g)}$ ions are $-406\,kJ\,mol^{-1}$ and $-364\,kJ\,mol^{-1}$. The enthalpy of solution of $NaCl_{(s)}$ is.....$kJ\,mol^{-1}$

(A) 738 (B) 10

(C) -10 (D) -822

(143) Oxidation number of As atoms in H_3AsO_4 is

(A) -3 (B) $+4$

(C) $+6$ (D) $+5$

(144) Octane number has 0 value for

(A) iso-octane (B) $n-$ hexane

(C) $n-$ heptane (D) iso-heptane

(145) If degree of ionisation is 0.01 of decimolar solution of weak acid HA then pKa of acid is

(A) 5 (B) 7

(C) 2 (D) 3

(146) What is major product of following reaction?

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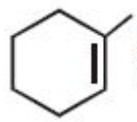
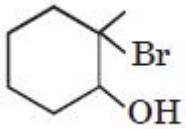
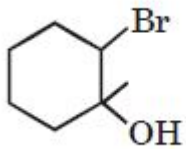
$\xrightarrow[H_2O]{Br_2}$



(A)



(B)



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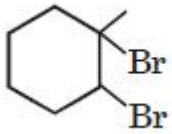
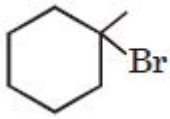


(D)



(147) $\underset{\underset{OH}{|}}{Me_2CH-CH-Me} \xrightarrow[350\,^oC]{Al_2O_3} (A) \xrightarrow[(ii)\,AgOH]{(i)\,HI} (B)$

Product (B) of above reaction

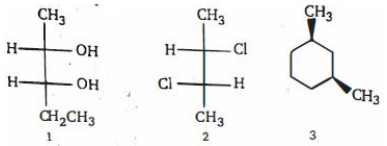
(A) $Me_2C(OH)CH_2Me$

(B) $\underset{\underset{OH}{|}}{Me_2CH-CH-Me}$

(C) $\underset{\underset{OH}{|}}{Me-CH-CMe_3}$

(D) $HO-CH_2-(CH_2)Me$

(148) Which of the following compounds are meso forms ?



(A) 1 only (B) 3 only

(C) 1 and 2 (D) 2 and 3

(149) The $IUPAC$ name of the compound $CH_2=CH-CH(CH_3)_2$ is

(A) $1,1-$ dimethyl $-2-$ propene

(B) $2-$ vinyl propene

(C) $3-$ methyl but $-1-$ ene

(D) $1-$ isopropyl ethene

(150) What is prefix of ester ?

(A) Oate (B) Haloformyl

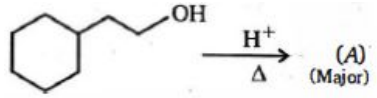
(C) Carbalkoxy (D) Formyl

(151) Which of the following is most reactive towards nucleophilic attack

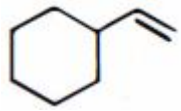
(A) $\underset{\underset{O}{||}}{CH_3-C-OC_2H_5}$ (B) $\underset{\underset{O}{||}}{CH_3-C-Cl}$

(C) $\underset{\underset{OO}{||||}}{CH_3-C-O-C-CH_3}$ (D) $\underset{\underset{O}{||}}{CH_3-C-NMe_2}$

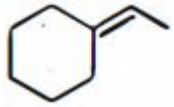
(152) Major-product (A) is:



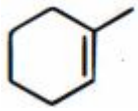
(A)



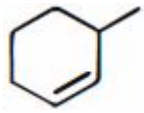
(B)



(C)



(D)



(153) The number of all types of isomers of chlorobutane is

(A) 2 (B) 4

(C) 6 (D) 5

(154) If H^+ ion concentration of a solution is increased by 10 times its pH will be

(A) Increase by one (B) Remains unchanged

(C) Decrease by one (D) Increase by 10

(155) In graphite $C-$atom is in ....state

(A) sp^3 (B) sp

(C) sp^2 (D) None of these

(156) For the following gaseous reaction $H_2+I_2 \rightleftharpoons 2HI$, the equilibrium constant

(A) $K_p > K_c$ (B) $K_p < K_c$

(C) $K_p = K_c$ (D) $K_p = 1/K_c$

(157) Find ....kJ/mol enthalpy of neutralisation of NH_4OH and HCN in aqueous solution if enthalpy of ionisation of NH_4OH and HCN are 7 kJ/mol and 8 kJ/mol. also enthalpy of ionisation of H_2O is 57.3 $kJ/mole$

(A) -15 (B) -42.3

(C) $+1$ (D) 42.3

(158) process $N_{2(g)}+3H_{2(g)} \rightleftharpoons 2NH_{3(g)}+heat$

(A) $K_p = K_c$ (B) $K_p = K_c\,(RT)^{-1}$

(C) $K_p = K_c\,(RT)^{-2}$ (D) $K_p = K_c\,(RT)$

(159) Fischer Tropsch process is used for the manufacture of

(A) Synthetic petrol (B) Thermosetting plastics

(C) Ethanol (D) Benzene

(160) The reaction $CH_3-\underset{\underset{OH}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3 \xrightarrow{H_2SO_4} CH_3-\overset{\overset{CH_3}{|}}{C}=CH_2$ is the example of

(A) Sulphonation (B) Dehydration

(C) Alkylation (D) Decomposition

(161) Which one of the following series contain's electrophiles only ?

(A) $H_2O,\ SO_2,\ H_3O^{\oplus}$ (B) $NH_3,\ H_2O,\ AlCl_3$

(C) $AlCl_3,\ SO_3,\ \overset{\oplus}{N}O_2$ (D) $H_2O,\ C\overset{\oplus}{l},\ NH_3$

(162) What is the weight of oxygen that is required for the complete combustion of 2.8 kg of ethylene? ....kg

(A) 9.6 (B) 96

(C) 6.4 (D) 2.8

(163) Which of the following compounds which is an optically active compound

(A) $1-$butanol (B) $2-$butanol

(C) $3-$butanol (D) $4-$heptanol

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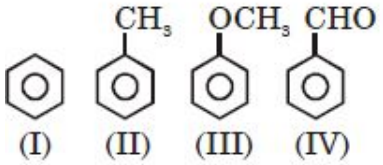
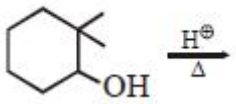
(164) Increasing order of electrophilic substitution for following compounds



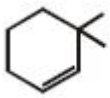
(A) $IV < I < II < III$ (B) $III < I < II < IV$

(C) $I < IV < III < II$ (D) $II < III < I < IV$

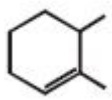
(165) Major product is



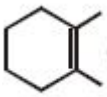
(A)



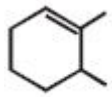
(B)



(C)



(D)

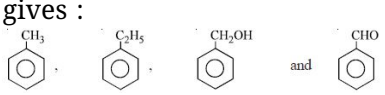


(166) Consider the following reaction
$C_6H_6(l) + \frac{15}{2}O_2(g) \rightarrow 6CO_2(g) + 3H_2O(g)$
Signs of ΔH, ΔS and ΔG for the above reaction will be

(A) $+, -, +$ (B) $-, +, -$

(C) $-, +, +$ (D) $+, +, -$

(167) If the mixture of the following four aromatic compounds on oxidation by strong oxidising agent gives :



(A) Mixture of $C_6H_5CH_2OH + C_6H_5COOH$

(B) Mixture of $C_6H_5CHO + C_6H_5COOH$

(C) Only C_6H_5COOH

(D) None of the above

(168) Which of the following compounds are not arranged in order of decreasing reactivity towards electrophilic substitution

(A) Fluoro benzene > chloro benzene > bromo benzene

(B) Phenol > $n-$propyl benzene > benzoic acid

(C) Chloro toluene > para-nitro toluene > $2-$chloro$-4-$nitro toluene

(D) Benzoic acid > phenol > $n-$propyl benzene

(169) How many carbon atoms in the molecule $HCOO-(CHOH)_2-COOH$ are asymmetric

(A) 1 (B) 2

(C) 3 (D) None of these

(170) Which compound has the highest value of pK_a ?

(A) $Cl-CH_2-CH_2-COOH$

(B) CH_3-CH_2-COOH

(C) $CH_3-\underset{\underset{Cl}{|}}{CH}-COOH$

(D) $CH_3-\overset{\overset{Cl}{|}}{\underset{\underset{Cl}{|}}{C}}-COOH$

(171) Hybridisation of carbon C_2 and C_4 respectively are $CH_3-CH=CH-CH_3$

(A) sp^2 and sp (B) sp and sp^2

(C) sp^3 and sp^2 (D) sp^2 and sp^3

(172) Assign A, B, C, D from given type of reaction.

$$NaNO_2 + FeSO_4 + H_2SO_4 \longrightarrow \underset{\text{(Ring complex)}}{[Fe(H_2O)_5NO]\,SO_4}$$

(A) for disproportionation reaction.

(B) for comproportionation reaction.

(C) for either intermolecular redox reaction or displacement reaction.

(D) for either thermal combination redox reaction or thermal decomposition redox reaction.

(173) In the process $BCl_3 + PH_3 \rightarrow BCl_3 : PH_3$ The Lewis acid is

(A) BCl_3 (B) PH_3

(C) Both (D) None

(174) For the equilibrium
$SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$, what is the temperature at which $\frac{K_p(atm)}{K_c(M)} = \frac{1}{3}$? .....K

(A) 0.027 (B) 0.36

(C) 36.54 (D) 4.06

(175) For the reaction $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$, true condition is

(A) $K_p = K_c$ (B) $K_p > K_c$

(C) $K_p < K_c$ (D) $K_c = 0$ but $K_p \neq 0$

(176) H_2O_2 is used as

(A) An oxidant only

(B) A reductant only

(C) An acid only

(D) An oxidant, a reductant and an acid

(177) What is the correct $IUPAC$ name for

$$CH_3-\overset{\overset{H}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH=CH-CH_2-\overset{\overset{O}{||}}{C}-OH$$

(A) $5-$ methyl $-3-$ hexenoic acid

(B) $5-$ carboxyl $-2-$ methylpentene

(C) $4-$ isopropyl $-3-$ butenoic acid

(D) None of above

(178) How many millilitres of $6.0\ M$ hydrochloric acid should be used to prepare $150\ ml$ of a solution which is $0.30\ M$ in hydrogen ion

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(A) 3 (B) 7.5

(C) 9.3 (D) 30

(179) Which one of the following reactions is not an example of redox reaction

(A) $Cl_2 + 2H_2O + SO_2 \rightarrow 4H^+ + SO^{4-} + 2Cl^-$

(B) $Cu^{++} + Zn \rightarrow Zn^{++} + Cu$

(C) $2H_2 + O_2 \rightarrow 2H_2O$

(D) $HCl + H_2O \rightarrow H_3O^+ + Cl^-$

(180) The equilibrium concentration of X, Y and YX_2 are 4, 2 and 2 moles respectively for the equilibrium $2X + Y \rightleftharpoons YX_2$. The value of K_c is

(A) 0.625 (B) 0.0625

(C) 6.25 (D) 0.00625

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Bright Tuition classs

Subject : Biology, Physics, Chemistry
Standard : 11
Total Mark : 720

NEET practice 15 th march

(Answer Key)

Paper Set : 1
Date : 15-03-2026
Time : 3H:0M

Biology - Section A (MCQ)

1 - A	2 - C	3 - C	4 - A	5 - C	6 - C	7 - B	8 - D	9 - D	10 - C
11 - D	12 - D	13 - D	14 - C	15 - C	16 - C	17 - A	18 - B	19 - B	20 - B
21 - B	22 - B	23 - C	24 - A	25 - C	26 - B	27 - A	28 - C	29 - C	30 - D
31 - C	32 - A	33 - A	34 - D	35 - B	36 - D	37 - D	38 - C	39 - B	40 - C
41 - C	42 - C	43 - C	44 - C	45 - A	46 - B	47 - B	48 - D	49 - D	50 - B
51 - C	52 - A	53 - D	54 - D	55 - D	56 - D	57 - C	58 - A	59 - A	60 - A
61 - D	62 - B	63 - C	64 - D	65 - A	66 - D	67 - C	68 - B	69 - A	70 - C
71 - D	72 - C	73 - A	74 - A	75 - A	76 - A	77 - D	78 - B	79 - A	80 - D
81 - B	82 - B	83 - B	84 - D	85 - A	86 - B	87 - B	88 - B	89 - A	90 - A

Physics - Section A (MCQ)

91 - D	92 - B	93 - A	94 - B	95 - A	96 - C	97 - A	98 - D	99 - C	100 - D
101 - D	102 - A	103 - A	104 - D	105 - C	106 - C	107 - D	108 - A	109 - C	110 - C
111 - A	112 - C	113 - D	114 - D	115 - C	116 - B	117 - B	118 - C	119 - A	120 - D
121 - B	122 - B	123 - A	124 - D	125 - C	126 - A	127 - C	128 - A	129 - A	130 - B
131 - D	132 - D	133 - D	134 - D	135 - C					

Chemistry - Section A (MCQ)

136 - B	137 - C	138 - B	139 - C	140 - D	141 - A	142 - B	143 - D	144 - C	145 - A
146 - B	147 - A	148 - B	149 - C	150 - C	151 - B	152 - B	153 - D	154 - C	155 - C
156 - C	157 - B	158 - C	159 - A	160 - B	161 - C	162 - A	163 - B	164 - A	165 - C
166 - B	167 - C	168 - D	169 - B	170 - B	171 - D	172 - C	173 - B	174 - D	175 - C
176 - D	177 - A	178 - B	179 - D	180 - B					

Bright Tuition class

Bright Tuition classs

Subject : Biology, Physics, Chemistry
Standard : 11
Total Mark : 720

NEET practice 15 th march

(Solutions)

Paper Set : 1
Date : 15-03-2026
Time : 3H:0M

Biology - Section A (MCQ)

(1) Black rust of wheat is caused by

(A) Puccinia graminis (B) Ustilago
(C) Pythium (D) None of these

Solution:(Correct Answer:A)
(a)Puccinia graminis tritici causes black rust of wheat.

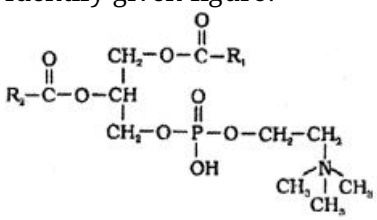
(2) A : Chitinous exoskeleton is a characteristic feature of arthropods.

R : It allows diffusion of water vapour from atmosphere to the body.

(A) Assertion and Reason both are correct and also correct explanation.
(B) Assertion and Reason both are correct but not explanation of assertion.
(C) Assertion is correct, but Reason is incorrect.
(D) Both Assertion and Reason are incorrect.

Solution:(Correct Answer:C)
Assertion is true, Chitinous exoskeleton is a characteristic feature of arthropods. Reason is false as chitin is impermeable to water and does not allow diffusion of water vapour from atmosphere to body or vice-versa. Assertion is true but Reason is false.

(3) Identify given figure.



(A) Cholesterol (B) Triglycerides
(C) Lecithin (D) Adinylic acid

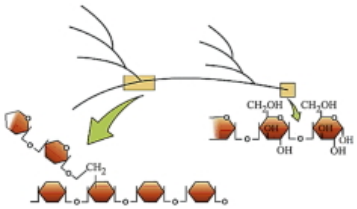
Solution:(Correct Answer:C)

(4) Which of the following respires by gills

(A) Prawn (B) Frog
(C) Crocodile (D) Whale

Solution:(Correct Answer:A)
It's obvious

(5) Which one is not correct for given structure of glycogen.



(A) There is only one reducing end
(B) Total six a$1-6$ glycosidic bond are present
(C) $\alpha 1-6$ glycosidic bond is more than $\alpha 1-4$ glycosidic bond
(D) each branch end with Non reducing end.

Solution:(Correct Answer:C)

(6) Choose correct pair.

$Column-I$	$Column-II$
$1.Phaeophyceae$	$A.Chara$
$2.Chlorophyceae$	$B.Wolffia$
$3.Rhodophyceae$	$C.Porphyra$
$4.Angiosperms$	$D.Ectocarpus$

(A) $(1-D),(2-A),(3-B),(4-C)$
(B) $(1-D),(2-C),(3-B),(4-A)$
(C) $(1-D),(2-A),(3-C),(4-B)$
(D) $(1-C),(2-A),(3-D),(4-B)$

Solution:(Correct Answer:C)

(7) Which of the following $statement(s)$ is/are correct for Metabolism?

(A) It is the sum total of all physical reactions taking place inside a living system.
(B) All plants, animals, fungi and microbes exhibit metabolism.
(C) Isolated metabolic reactions in-vitro are not living but are living reactions.
(D) All of the above

Solution:(Correct Answer:B)
Metabolism is defined as the sum total of all the chemical reactions occurring in our body. All plants, animals, fungi and microbes exhibit metabolism. Isolated metabolic reactions in vitro are not living things but are nonliving reactions.

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(8) Read the following statement and choose the correct characteristic feature of Aves

(A) Skin of birds have glands

(B) Air sacs help in excretion

(C) Heart is having three auricle and one ventricle

(D) Preen gland is present at the base of tail

Solution:(Correct Answer:D)
Skin in Aves is dry and without glands, except oil gland and spleen gland at base of tail for lubrication of feathers. Air sacs in Aves help in respiration and excretory function in Aves is mediated by kidneys. Heart is completely four chambered in birds.

(9) Growth in living organisms occurs by

(A) division of cells.

(B) increase in biomass.

(C) accumulation of materrial by external agency.

(D) both (a) and (b).

Solution:(Correct Answer:D)
Accumulation of material by external agency cause extrinsic growth which can not be the feature of living organism.

(10) Which of the following mode of respiration always continue in frog?

(A) buccopharyngeal (B) Pulmonary

(C) Cutaneous (D) Gills

Solution:(Correct Answer:C)

(11) Sexual reproduction in algae results in the formation of

(A) Oospore (B) Zoospore

(C) Zygote **(D)** Zygospore

Solution:(Correct Answer:D)
(d) Fusion of gametes results in the formation of zygote, which by secreting a thick wall acts as perennating spore $(zygospore)$ and helps in perennation, dispersal and multiplication.

(12) Microbodies differ from lysosomes in that

(A) Microbodies are surrounded by a single unit membrane while lysosome membrane is double

(B) Microbodies are surrounded by double membrane while lysosomes membrane is single unit

(C) Microbodies contain lytic enzymes while lysosomes do not

(D) Lysosome contain lytic enzymes while microbodies do not

Solution:(Correct Answer:D)
It's obvious.

(13) Bryophytes resemble algae in the following aspect.

(A) Filamentous body, presence of vascular tissues and autotrophic nutrition

(B) Differentiation of plant body into root, stem and leaves and autotrophic nutrition

(C) Thallus like plant body, presence of roots and autotrophic nutrition

(D) Thallus like plant body, lack of vascular tissues and autotrophic nutrition

Solution:(Correct Answer:D)
Bryophytes resemble algae in many ways, some of which are (i) thalloid plant body (ii) absence of roots (iii) absence of complex vascular tissues (iv) autotrophic mode of nutrition (v) reserve food material is true starch.

(14) A nucleotide has three chemically distinct compounds. These are A, B and C

(A) A - Sugar, B-carbonates, C-chlorides

(B) A-DNA, B-cellulose, C-chitin

(C) A-Purine or pyrimidine B-Monosaccharide, C-Phosphate

(D) A-Phosphoric acid, B-Proteins, C-acids

Solution:(Correct Answer:C)

(15) It is not part of flower.

(A) Gynoecium (B) Androecium

(C) Bract (D) Corolla

Solution:(Correct Answer:C)

(16) Branchiostoma is an example of

(A) Cyclostomata (B) Chondrichthyes

(C) Cephalochordata (D) Osteichthyes

Solution:(Correct Answer:C)

(17) Vegetative reproduction by fragmentation is common in

(A) Agaricus (B) Saccharomyces

(C) Euglena (D) Gonyaulax

Solution:(Correct Answer:A)
Vegetative reproduction by fragmentation is common in fungi.

(18) The cell wall is impermeable to water and deposition of suberin is also found in

(A) Bast **(B)** Cork

(C) Bark (D) Xylem

Solution:(Correct Answer:B)
It's obvious.

(19) Choose correct pair.

(A) Nitrogen base - alanine

(B) Polysaccharide - glycogen

(C) Amino acid - trypsin

(D) Enzyme - insulin

Solution:(Correct Answer:B)

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(20) Identify the plant parts whose transverse section shows a clear and prominent pith.

(A) Dicot and monocot stems

(B) Dicot stem and monocot root

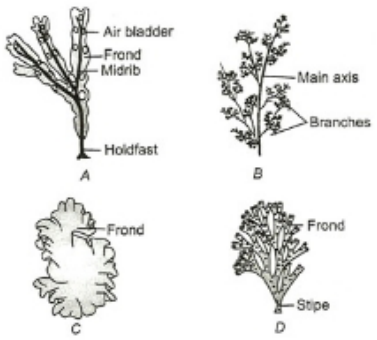
(C) Dicot and monocot roots

(D) Dicot stem and dicot root

Solution:(Correct Answer:B)

A thin-walled pith is generally present in monocot roots, while in dicot roots, a thin-walled conjunctive tissue is present in between vascular elements. Thinwalled pith is also well marked in dicot stems but absent in monocot stems.

(21) Identify the given figures of algae and select the correct option



(A) $A - Volvox, B - Chlamydomonas, C - Chara, D - Porphyra$

(B) $A - Fucus, B - Polysiphonia, C - Porphyra, D - Dictyota$

(C) $A - Fucus, B - Dictyota, C - Porphyra, D - Polysiphonia$

(D) $A - Dictyota, B - Porphyra, C - Fucus, D - Polysiphonia$

Solution:(Correct Answer:B)

null

(22) Match the following column I with coloumn II and choose the correct combination.

Column $-I$	Column $-II$
(a) Earthworm	(i) Gizzard
(b) Cockroach	(ii) Caecum
(c) Frog	(iii) Clitellum
(d) Rat	(iv) Cloaca

(A) $(a) - (i), (b) - (ii), (c) - (iv), (d) - (iii)$

(B) $(a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)$

(C) $(a) - (iii), (b) - (i), (c) - (ii), (d) - (iv)$

(D) $(a) - (i), (b) - (iii), (c) - (iv), (d) - (ii)$

Solution:(Correct Answer:B)

It's Obvious

(23) When fungi feed on dead organic matter, they are known as

(A) Dimorphic (B) Parasites

(C) Saprophytes (D) None of these

Solution:(Correct Answer:C)

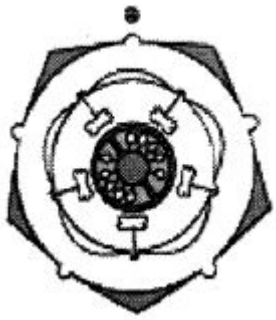
(c)All fungi are heterotrophs in their nutrition. Some are depends upon organic matter known as saprophytes.

(24) Select the correct floral diagram of Brassicaceae

(A)

(B)

(C)



(D)

Solution:(Correct Answer:A)

(25) In which the pneumatophores are found

(A) Tinospora (B) Pinus

(C) Rhizophora (D) None of these

Solution:(Correct Answer:C)

(c) Pneumatophores develop from horizontal roots. They bear a number of lenticels or pneumathodes. Exchange of gases occurs through these pores. e.g., Rhizophora.

(26) Stems are modified into flattened structures, which carryout photosynthesis in

(A) Euphorbia **(B)** Opuntia

(C) Bougainvillea (D) Colocasia

Solution:(Correct Answer:B)

In Opuntia, stems are modified into flattened structures which carryout photosynthesis.

(27) Identify the correct option for Cyclostomata : –

(A) Development indirect, $6 - 15$ pairs of gillslits, ectoparasites on some fishes

(B) Development indirect, viviparous, $6 - 15$ pairs of gill–slits, scales and paired fins present.

(C) Jawless, scales absent, only paired fins are present

(D) Both paired and unpaired fins are present, ventral mouth, $6 - 15$ pairs of gill–slits

Solution:(Correct Answer:A)

(28) Which of the following is not a significance of scientific nomenclature?

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(A) It ensure that one organism is recognised by one scientific name only.

(B) One scientific name is given to one organism only.

(C) Description of one organism leads to two scientific name all around the world.

(D) 1 and 2 both

Solution:(Correct Answer:C)

(29) Systematics refers to the

(A) identification and classification of plants and animals.

(B) nomenclature and identification of plants and animals.

(C) diversity of kinds of organisms and their relationship.

(D) different kinds of organisms and their classification.

Solution:(Correct Answer:C)
Systematics, often used interchangeably with taxonomy, is the study of diversity of organisms, their comparative and evolutionary relationships on the basis of findings from various fields of biology.

(30) Aerenchyma is helpful in plants by

(A) Giving flexibility to plants

(B) Giving mechanical strength to plants

(C) Promoting photosynthesis

(D) Providing buoyancy in hydrophytes

Solution:(Correct Answer:D)
(d) In parenchyma of hydrophytes, large air cavities are formed. Due to the presence of air the plants become buoyant.

(31) Ornithorhynchus is a

(A) A fossil bird (B) Duck

(C) Monotreme mammal (D) Dinosaur

Solution:(Correct Answer:C)
It's obvious

(32) I. Made up of sclerenchymatous cells

II. Generally absent in primary phloem

III. Much elongated and pointed

Given above characters belongs to which of the following?

(A) Phloem fibre (B) Xylem fibre

(C) Companion cells (D) Sieve cells

Solution:(Correct Answer:A)
Phloem fibres (bastfibres) are made up of sclerenchymatous cells. These are generally absent in the primary phloem but are found in the secondary phloem. These are much elongated, unbranched and have pointed, needle like apices. The cell wall of phloem fibres is quite thick. At maturity, these fibres lose their protoplasm and become dead. Phloem fibres of jute, flax and hemp are used commercially

(33) The roots of gymnosperms is

(A) Tap root (B) Adventitious roots

(C) Rhizoid (D) Rhizophores

Solution:(Correct Answer:A)
It's obvious.

(34) A : The inner lining of trachea, bronchi and fallopian tubes has simple epithelium with non-motile cilia.

R : Cilia form the brush-border.

(A) Assertion and Reason both are correct and also correct explanation.

(B) Assertion and Reason both are correct but not explanation of assertion.

(C) Assertion is correct, but Reason is incorrect.

(D) Both Assertion and Reason are incorrect.

Solution:(Correct Answer:D)
Assertion is false as epithelium lining inner surface of hollow organs like fallopian tubes and most of respiratory tract is ciliated columnar cells. Reason is also false as brush bordered epithelium of intestinal mucosa bear microvilli on surface. Both Assertion and Reason is false.

(35) Ribosome mainly has

(A) DNA **(B)** RNA

(C) Carbohydrate (D) None of these

Solution:(Correct Answer:B)
It's obvious.

(36) Zygospores are formed in

(A) Puccinia (B) Penicillium

(C) Alternaria **(D)** Mucor / Rhizopus

Solution:(Correct Answer:D)
(d) Fusion of coenogametes $(Rhizopus\, and\, Mucor)$ produced a diploid resting spore called zygospore.

(37) The practical purpose of classification of living organisms is to

(A) explain the origin of living organsims.

(B) trace the evolution of living organsims.

(C) name the living organisms.

(D) facilitate identification of unknown organisms.

Solution:(Correct Answer:D)
Biological classification is the scientific arrangement of organisms in a hierarchial series of groups and subgroups on the basis of similarities and differences in their traits. It helps in building evolutionary pathways and in identifying new organisms.

(38) Which of the following pairs come under the group chrysophytes?

(A) Diatoms and Euglena

(B) Euglena and Trypanosoma

(C) Diatoms and Desmids

(D) Gonyaulax and Desmids

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Solution:(Correct Answer:C)
Chrysophytes are a group of diatoms, golden algae (desmids) and golden brown photosynthetic microscopic protists. Their body is covered by a transparent siliceous shell.

(39) Roots are used in vegetative propagation in
(A) Potato **(B)** Sweet potato
(C) Ginger (D) Onion

Solution:(Correct Answer:B)
(b)Reproductive roots are present in sweet potato. The adventitious roots may develop adventitious buds and help in vegetative propagation.

(40) Largest phylum, jointed appendages, chitinous exoskeleton, mostly dioecious The above features belongs to : –
(A) Pheretima (B) Fasciola
(C) Apis (D) Ascaris

Solution:(Correct Answer:C)

(41) Which is not an example of prokaryotic cell ?
(A) Bacteria (B) Mycoplasma
(C) Egg of ostrich (D) $PPLO$

Solution:(Correct Answer:C)

(42) In Gracilaria, sexual reproduction is
(A) Isogamous (B) Anisogamous
(C) Oogamous (D) Both (a) and (b)

Solution:(Correct Answer:C)
Sexual reproduction in red algae is oogamous e.g., Gracilaria, Gelidium

(43) Given organisms belongs to how many genera?

Wheat, Brinjal, Potato, Lion, Dog, Tiger

(A) Three (B) Two
(C) Four (D) Five

Solution:(Correct Answer:C)
Animal Genera
Wheat -Triticum
Brinjal & Potato -Solanum $\Rightarrow$ Four genera
Lion & Tiger -Panthera
Dog -Can

(44) Epipetalous androecium is found in
(A) Mung (B) Pea
(C) Solanum nigrum (D) Groundnut

Solution:(Correct Answer:C)

(45) Stomata are the component of
(A) Epidermal tissue system
(B) Ground tissue system
(C) Conducting tissue system
(D) Vascular tissue system

Solution:(Correct Answer:A)
Epidermal tissue system includes $\rightarrow$ Epidermis, stomata, lenticels, trichomes
While ground tissue system includes all tissues excluding vascular tissue system and epidermal tissue system.
Vascular tissue system includes xylem and phloem.

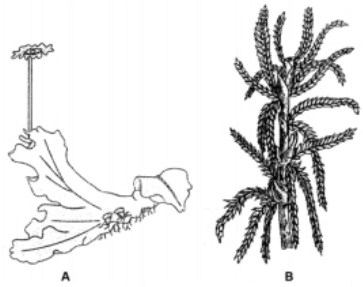
(46) The structure of nuclear membrane facilitates
(A) Synapsis of homologous chromosomes at meiosis
(B) Nucleo-cytoplasmic exchange of materials
(C) Anaphasic separation of daughter chromosomes
(D) Organization of spindles

Solution:(Correct Answer:B)
It's obvious.

(47) DNA is
(A) Always double stranded
(B) Rarely single stranded
(C) Always single stranded
(D) Rarely double stranded

Solution:(Correct Answer:B)
It's Obvious

(48) Identify the plants A and B in the figures given below



(A) A - Female Marchantia, B - Sphagnum
(B) A - Female Marchantia, B - Sphagnum
(C) A - Marchantia, B - Funaria
(D) A - Male Marchantia, B - Sphagnum

Solution:(Correct Answer:D)
Plant 'A' -Male Marchantia Plant 'B' -Sphagnum

(49) Which of the following is incorrect?
(A) Mycoplasma is the smallest cell
(B) Bacteria are 3 to 5 μm in size.
(C) The largest cell is the egg of an ostrich.
(D) Nerve cells are some of the smallest cells.

Solution:(Correct Answer:D)
Nerve cells are the longest cells of the body.

(50) Dominant protein in biosphere is
(A) Trypsin **(B)** Rubisco
(C) Collagen (D) Keratin

Solution:(Correct Answer:B)

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(51) Match the following:

Column - I	Column - II
1. Smooth muscles	p. Hard and not pliable ground substance
2. Skeletal muscles	q. Blood vessel stomach intestine
3. Bone	r. Condroblast
4. Cartilage	s. A-band

(A) $(1-p),(2-q),(3-r),(4-s)$

(B) $(1-q),(2-p),(3-s),(4-r)$

(C) $(1-q),(2-s),(3-p),(4-r)$

(D) $(1-p),(2-r),(3-s),(4-q)$

Solution:(Correct Answer:C)

(52) Vessels are mainly found in

(A) Xylem of angiosperms (B) Xylem of gymnosperms

(C) Both (a) and (b) (D) None of these

Solution:(Correct Answer:A)
It's obvious.

(53) The calyptrogen of the root apex forms

(A) Rhizoids (B) Root nodule

(C) Root hairs **(D)** Root cap

Solution:(Correct Answer:D)
(d) Root cap : In root apex, calyptrogen is covered by a cap like structure known as root cap, which helps the root apex from friction of soil.

(54) Mark the odd one is each series :
(a) Fat body, Nephrocyte, Urinary bladder (in cockroach)
(b) Blood, Bone, Cartilage, Muscle
(c) Skeletal, Cardiac, Nerve, Smooth Muscles

(A) $(a)-$Nephrocyte, $(b)-$Muscle, $(c)-$Nerve

(B) $(a)-$Fat body, $(b)-$Cartilage, $(c)-$Cardiac Muscle

(C) $(a)-$Nephrocyte, $(b)-$Bone, $(c)-$Skeletal Muscle

(D) $(a)-$Urinary Bladder, $(b)-$Muscle, $(c)-$Nerve

Solution:(Correct Answer:D)

(55) Chlorophyll containing fleshy cylindrical structures found in Euphorbia are modified

(A) Roots (B) Fruit

(C) Leaves **(D)** Stem

Solution:(Correct Answer:D)
Phylloclade is stem modification in Euphorbia.

(56) An example of agarophyte is

(A) Dictyota (B) Fucus

(C) Nostoc **(D)** Gelidium

Solution:(Correct Answer:D)
(d) Gelidium is a red algae which produces agar-agar.

(57) Which covers the largest number of organisms

(A) Genus (B) Family

(C) Phylum (D) Class

Solution:(Correct Answer:C)
(c) In phylum included one or more related classes. All the classes of a phylum possess some common correlated traits. So the phylum covers the largest numbers of organism.

(58) Archaebacteria found in extremely salty area

(A) Halophiles (B) Methanogens

(C) Thermoacidophiles (D) Cyanobacteria

Solution:(Correct Answer:A)

(59) Starch and cellulose are the compounds made up of many units of

(A) Simple sugar (B) Fatty acid

(C) Glycerol (D) Amino acid

Solution:(Correct Answer:A)
(a) Simple sugar $i.e.$ monosaccharide. These are the simplest carbohydrates and are building units of complex carbohydrates. $i.e.$ Starch and cellulose.

(60) Choose the correct names of the different bacteria given below according to their shapes.

(A) A- Cocci, B - Bacilli, C - Spirilla, D - Vibrio

(B) A - Bacilli, B- Cocci, C - Spirilla, D - Vibrio

(C) $A-$ Spirilla, B - Bacilli, C - Cocci, D - Vibrio

(D) $A-$ Spirilla, B - Vibrio, C- Cocci, D - Bacilli

Solution:(Correct Answer:A)
A - Cocci (spherical), B - Bacilli (rod shaped), C Spirilla (spiral), D - Vibrio (comma shaped). These are all the shapes of the bacteria.

(61) Non poisonous snake is : $-$

(A) Vipera (B) Naja

(C) Bangarus **(D)** Azgar

Solution:(Correct Answer:D)

(62) How many statements are correct?
(A) Animal cells contain membrane bound organelle called centriole which help in cell division.
(B) Lipids are arranged within the plasma membrane with the polar head towards the inner sides and the hydrophobic tails towards the outer side.
(C) The cell wall and middle lamellae may be traversed by plasmodesmata which connect the cytoplasm of neighboring plant cells.
(D) The quasifluid nature of lipid enables lateral movement of proteins within the overall bilayer.

(A) 1 **(B)** 2

(C) 3 (D) 4

Solution:(Correct Answer:B)

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(63) What is correct for schlerenchyma tissue

(A) Intercellular space, oval shaped cell, photosynthesis, storage and secretion are functions of its

(B) Cells may be oval, spherical or polygonal and contain chlorophyll, living cells.

(C) Dead cells and without protoplast, provides mechanical support

(D) Cell wall is thin and made up of cellulose, intercellular space is present

Solution:(Correct Answer:C)

(64) Fishes, amphibians, reptiles and birds are kept in the same

(A) Order (B) Class

(C) Genus **(D)** Phylum

Solution:(Correct Answer:D)

null

Animals	Class	Phylum
Fishes	Pisces	Chordata
Amphibians	Amphibians	Chordata
Reptiles	Reptilia	Chordata
Birds	Aves	Chordata

(65) Stilt roots are reported from

(A) Pandanus (B) Radish

(C) Mango-ginger (D) Bryophyllum

Solution:(Correct Answer:A)

(a) In some plants roots arise from lower nodes of stem and enter the soil to provide extrasupport. They behave as ropes of a tent. Such roots are called stilt or brace roots e.g., Pandanus.

(66) Identify the correct statement regarding enzymes

(A) Enzymes are specific biological catalysts that can normally function at very high temperatures ($T1000\ K$)

(B) Enzymes are normally heterogeneous catalysts that are very specific in their action

(C) Enzymes are specific biological catalysts that cannot be poisoned

(D) Enzymes are specific biological catalysts that possess well defined active sites

Solution:(Correct Answer:D)

Enzymes have well defined active sites and their action are specific in nature. They are called biological catalysts and work at optimum temperature between $25°C$ to $40°C$

(67) Frog which lives on the trees

(A) Alytes (B) Bufo

(C) Hyla (D) Rana

Solution:(Correct Answer:C)

(c)Hyla is also known as tree frog.

(68) Intercalary meristem results in

(A) Secondary growth **(B)** Primary growth

(C) Apical growth (D) Periderm formation

Solution:(Correct Answer:B)

It's obvious.

(69) Scientific name of wheat.

(A) Triticum aestivum (B) Anacardiaceae

(C) Musca domestica (D) Mangifera indica

Solution:(Correct Answer:A)

(70) What are the three basic components of connective tissues?

(A) Ground substance, cells and basement membrane

(B) Cartilage, intercellular matrix and serum

(C) Cells, protein fibers and ground substance

(D) Collagen, elastin and reticular fibers

Solution:(Correct Answer:C)

Connective tissue fills the spaces between organs and tissues, and provides structural and metabolic support for other tissues and organs. The three basic components of connective tissue are cells, protein fibres and ground matrix.

(71) Which of the following is correct for middle lamella of eukaryotic cell ?

(A) It is formed in plant cell

(B) it is mainly consists of $Ca-$ pectate

(C) It holds different neighbouring cells together

(D) All of the above

Solution:(Correct Answer:D)

(72) Blind sac body plan is shown by

(A) Roundworms (B) Annelids

(C) Coelenterates (D) Arthropods

Solution:(Correct Answer:C)

(c)Blind sac body plan is found in coelenterates and flat worms. In this type of body plan animals has a single opening that act as both mouth and anus.

(73) Aestivation in cassia calotropis is respectively.....

(A) Imbricate, Valvate (B) Vexillary, twisted

(C) Valvate, Twisted (D) Vexillary, valvate

Solution:(Correct Answer:A)

(74) Identify the components labelled as A, B, C, D and E in the diagram given below from the list (i) to $(viii)$.

(i) Cristae of mitochondria (ii) Inner membrane of mitochondria (iii) Cytoplasm (iv) Smooth endoplasmic reticulum (v) Rough endoplasmic reticulum (vi) Mitochondrial matrix (vii) Ribosome $(viii)$ Nucleus

(A) $(viii)$ (v) (vii) (iii) (iv) (B) (i) (iv) (vii) (vi) (iii)

(C) (vi) (v) (iv) (vii) (i) (D) (v) (i) (iii) (ii) (iv)

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Solution:(Correct Answer:A)

In the given diagram of endoplasmic reticulum, the components marked as A, B, C, D and E are respectively nucleus $(viii)$, rough endoplasmic reticulum (v), ribosomes (vii), cytoplasm (iii) and smooth endoplasmic reticulum (iv). Endoplasmic reticulum is a network of membranous tubules within the cytoplasm of a eukaryotic cell, continuous with the nuclear membrane. It usually has ribosomes attached and is involved in protein and lipid synthesis.

(75) Mast cells are found in

(A) Connective tissue (B) Muscular tissue

(C) Nervous tissue (D) Blood

Solution:(Correct Answer:A)

It's Obvious

(76) Lacunae are connected with

(A) Canaliculae (B) Sublacunae

(C) Both (a) and (b) (D) None of the above

Solution:(Correct Answer:A)

(a) A lacunae gives off numerous fine branching tubules called canaliculi which connects to the lacunae from other.

(77) No cell could live without

(A) Phytochrome (B) Enzymes

(C) Chloroplasts **(D)** Protein

Solution:(Correct Answer:D)

(d) No cell could live without protein because proteins are building block of the body.

(78) Which components present in plasma membrane as result it having fluid structure?

(A) Carbohydrates **(B)** Lipid

(C) Glycoprotein (D) Polysaccherricles

Solution:(Correct Answer:B)

(79) Statement$-I$: Nuclear membrane is absent in bacteria.
Statement$-II$: Methanogens are responsible for the production of methane.

(A) Both statements I and II are correct

(B) Only statement I is incorrect

(C) Both statements I and II are incorrect

(D) Only statement II is incorrect

Solution:(Correct Answer:A)

(80)

Column-I (Terms)	Column-ll (Examples)
A. Ascus	I. Spirulina
B. Basidium	II. Penicillium
C. Protista	III. Agaricus
D. Cyanobacteria	IV. Euglena
E. Animalia	V. Sponges

Match the terms given in column-I with their examples given in column-II and choose the correct option

(A) $A-II, B-III, C-IV, D-V, E-I$

(B) $A-I, B-II, C-III, D-V, E-IV$

(C) $A-II, B-V, C-III, D-I, E-IV$

(D) $A-II, B-III, C-IV, D-I, E-V$

Solution:(Correct Answer:D)

Ascus is a sac, typically cylindrical in shape, in which the spores of ascomycetes fungi develop.eg Penicillinium,Aspergillus
Basidium is a microscopic club-shaped spore-bearing structure produced by club fungi.egAgaricus(Mushroom)
Euglena, Spirulina and sponges are the examples of protista, cyanobacteria and animalia respectively.

(81) In Cockroach, the number of ganglia are

(A) Two pairs thoracic and four pairs abdominal

(B) Three pairs thoracic and six pairs abdominal

(C) Three pairs thoracic and five pairs abdominal

(D) Two pairs thoracic and six pairs abdominal

Solution:(Correct Answer:B)

(b)The total number of ganglia in ventral nerve cord of cockroach is nine pairs, i.e. three pairs thoracic and six pairs adbominal.

(82) Coralloid roots have a symbiotic association with

(A) Photosynthetic green algae

(B) Nitrogen-fixing cyanobacteria

(C) Fungus

(D) Photosynthetic brown algae

Solution:(Correct Answer:B)

Coralloid roots have a symbiotic association with Nitrogen-fixing cyanobacteria.

(83) Which of the following statements regarding cyanobacteria is incorrect?

(A) It is also called blue green algae.

(B) They are chemosynthetic autotrophs.

(C) It forms blooms in polluted water bodies.

(D) It is unicellular, colonial or filamentous, marine or terrestrial bacteria.

Solution:(Correct Answer:B)

Cyanobacteria are photosynthetic (containing a blue photosynthetic pigment) autotrophs. They are prokaryotic and represent the earliest known form of life on the Earth. Certain cyanobacteria can fix atmospheric nitrogen and convert them to ammonia with the help of Heterocysts.

(84) Meristem is defined as a plant tissue where

(A) Cell conserves food and supply it to new ones

(B) Cells mature and add to the bulk of a plant

(C) Cells elongate and add to the growth of a plant

(D) Cells divide continuously to give rise to new ones

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Solution:(Correct Answer:D)
It's obvious.

(85) Which of the following stage of Funaria is haploid
(A) Gametophyte (B) Sporophyte
(C) Both (a) and (b) (D) None of these

Solution:(Correct Answer:A)
(a)Gametophyte stage of funaria is haploid because the main plant body is gametophyte.

(86) Layer of cells between endodermis and vascular bundles is called
(A) Epidermis **(B)** Pericycle
(C) Hypodermis (D) Pith

Solution:(Correct Answer:B)
(b) Pericycle is a single layered or multilayered cylinder of thin-walled or thick walled cells present between the endodermis and vascular tissues.

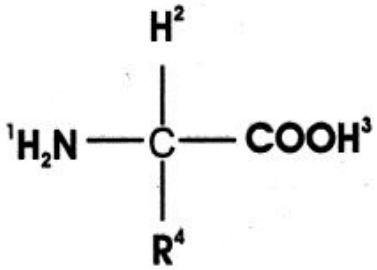
(87) Frog can protect itself from enemies by ?
(A) Teeth **(B)** Camouflage
(C) Clawed digit (D) Both (1) and (2)

Solution:(Correct Answer:B)

(88) When centromere present slightly away from the middle of the chromosome, then chromosome is called : –
(A) Metacentric chromosome
(B) Sub–metacentric chromosome
(C) Acrocentric chromosome
(D) Telocentric chromosome

Solution:(Correct Answer:B)

(89) Which of the two groups of following formula is involved in peptide bond formation between different amino acids?



(A) 1 and 3 (B) 2 and 4
(C) 2 and 3 (D) 1 and 4

Solution:(Correct Answer:A)

(90) ..... is family of cat.
(A) Felidae (B) Canidae
(C) Hominidae (D) Muscidae

Solution:(Correct Answer:A)

Physics - Section A (MCQ)

(91) Two forces $F_1 = 1\,N$ and $F_2 = 2\,N$ act along the lines $x = 0$ and $y = 0$ respectively. Then the resultant of forces would be
(A) $\hat{i} + 2\hat{j}$ (B) $\hat{i} + \hat{j}$
(C) $3\hat{i} + 2\hat{j}$ **(D)** $2\hat{i} + \hat{j}$

Solution:(Correct Answer:D)
(d) $x = 0$ means $y-$ axis $\Rightarrow$ $\vec{F}_1 = \hat{j}$
$y = 0$ means $x-$ axis $\Rightarrow$ $\vec{F}_2 = 2\hat{i}$
so resultant $\vec{F} = \vec{F}_1 + \vec{F}_2 = 2\hat{i} + \hat{j}$

(92) Two forces $\vec{F}_1 = 5\hat{i} + 10\hat{j} - 20\hat{k}$ and $\vec{F}_2 = 10\hat{i} - 5\hat{j} - 15\hat{k}$ act on a single point. The angle between $\vec{F}_1$ and $\vec{F}_2$ is nearly _____ $^\circ$.
(A) 30 **(B)** 45
(C) 60 (D) 90

Solution:(Correct Answer:B)
(B) The angle θ between two vectors is found using the dot product formula: $\cos\theta = \frac{\vec{F}_1 \cdot \vec{F}_2}{|\vec{F}_1||\vec{F}_2|}$.
$\vec{F}_1 \cdot \vec{F}_2 = (5 \times 10) + (10 \times -5) + (-20 \times -15) = 50 - 50 + 300 = 300$.
Magnitudes: $|\vec{F}_1| = \sqrt{5^2 + 10^2 + (-20)^2} = \sqrt{525}$ and $|\vec{F}_2| = \sqrt{10^2 + (-5)^2 + (-15)^2} = \sqrt{350}$.
$\cos\theta = \frac{300}{\sqrt{525} \times \sqrt{350}} \approx 0.707 = \frac{1}{\sqrt{2}}$.
Therefore, $\theta = 45^\circ$.

(93) A body A starts from rest with an acceleration a_1 . After $2\ seconds$, another body B starts from rest with an acceleration a_2 . If they travel equal distance in the $5th\ second$, after the start of A , then the ratio a_1 : a_2 is equal to
(A) $5:9$ (B) $5:7$
(C) $9:5$ (D) $9:7$

Solution:(Correct Answer:A)
According to problem
Distance travelled by body A in 5th sec and distance travelled by body B in 3rd sec. of its motion are equal.
$0 + \frac{a_1}{2}(2 \times 5 - 1) = 0 + \frac{a_2}{2}[2 \times 3 - 1]$
$9a_1 = 5a_2 \Rightarrow \frac{a_1}{a_2} = \frac{5}{9}$

(94) Wind is blowing in the north direction at speed of $2\ m/s$ which causes the rain to fall at some angle with the vertical. With what velocity should a cyclist drive so that the rain appears vertical to him :
(A) $2\ m/s$ south **(B)** $2\ m/s$ north
(C) $4\ m/s$ west (D) $4\ m/s$ south

Solution:(Correct Answer:B)
Horizontal component of rain's velocity will be equal to velocity of wind which is $2m/s$ in north direction. If cyclist goes towards north with velocity $2m/s$, then $w.r.t$ him rain's horizontal component of velocity will be zero, and he will see only vertical component.

(95) Two forces of $10\,N$ and $6\,N$ act upon a body. The direction of the forces are unknown. The resultant force on the body may be .....N
(A) 15 (B) 3
(C) 17 (D) 2

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Solution:(Correct Answer:A)

(a)
The resultant of two vectors always lie between $(A+B)$ and $(A-B)$. So the resultant of $10\,N$ and $6\,N$ should lie between $16\,N$ and $4\,N$.
So answer is $15\,N$.

(96) A body is moving in a circular path with a constant speed. It has

(A) A constant velocity

(B) A constant acceleration

(C) An acceleration of constant magnitude

(D) An acceleration which varies with time

Solution:(Correct Answer:C)

(c) Centripetal acceleration $= \frac{v^2}{r} =$ constant. Direction keeps changing.

(97) Given $\left|\vec{A}_1\right| = 2$, $\left|\vec{A}_2\right| = 3$ and $\left|\vec{A}_1 + \vec{A}_2\right| = 3$. Find the value or $\left|\left(\vec{A}_1 + 2\vec{A}_2\right) \times \left(3\vec{A}_1 - 4\vec{A}_2\right)\right|$

(A) 64 (B) 60

(C) 62 (D) 61

Solution:(Correct Answer:A)

(98) Which of the following is a scalar quantity

(A) Displacement (B) Electric field

(C) Acceleration **(D)** Work

Solution:(Correct Answer:D)

(d)Displacement, electrical and acceleration are vector quantities.

(99) A man projects a coin upwards from the gate of a uniformly moving train. The path of coin for the man will be

(A) Parabolic (B) Inclined straight line

(C) Vertical straight line (D) Horizontal straight line

Solution:(Correct Answer:C)

(c)Because horizontal velocity is same for coin and the observer. So relative horizontal displacement will be zero.

(100) If the length of the second's hand in a stop clock is $3\,cm$ the angular velocity and linear velocity of the tip is

(A) $0.2047\,rad/sec, 0.0314\,m/sec$

(B) $0.2547\,rad/sec, 0.314\,m/sec$

(C) $0.1472\,rad/sec, 0.06314\,m/sec$

(D) $0.1047\,rad/sec, 0.00314\,m/sec$

Solution:(Correct Answer:D)

(d) $\omega = \frac{2\pi}{T} = \frac{2\pi}{60} = 0.1047\ rad/s$
and $v = \omega r = 0.1047 \times 3 \times 10^{-2} = 0.00314\ m/s$

(101) Let $\vec{v}$ and $\vec{a}$ denote the velocity and acceleration respectively of a body in one-dimensional motion

(A) $|\vec{v}|$must decrease when $\vec{a} < 0$

(B) Speed will decrease when $|\vec{v}| < 0$ and $\vec{a}$ are < 0

(C) Speed will increase when both $|\vec{v}|$ and $\vec{a}$ are < 0

(D) Both (B) and (C)

Solution:(Correct Answer:D)

(102) For component of a vector $A = (3\hat{i} + 4\hat{j} - 5\hat{k})$, match the following colum.

Colum I	Colum II
(A) $x-$axis	(p) $5\,unit$
(B) Along another vector $(2\hat{i} + \hat{j} + 2\hat{k})$	(q) $4\,unit$
(C) Along $(6\hat{i} + 8\hat{j} - 10\hat{k})$	(r) 0
(D) Along another vector $(-3\hat{i} - 4\hat{j} + 5\hat{k})$	(s) None

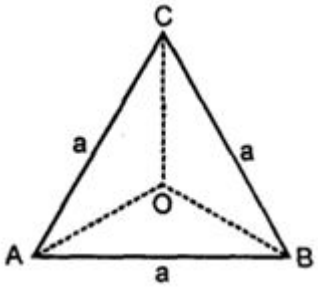
(A) $(A \to q, B \to r, C \to s, D \to s)$ (B) $(A \to p, B \to r, C \to s, D \to s)$

(C) $(A \to r, B \to q, C \to s, D \to s)$ (D) $(A \to q, B \to r, C \to s, D \to p)$

Solution:(Correct Answer:A)

(a)
$(2\hat{i} + \hat{j} + 4\hat{k})$ is perpendicular to A, because the dot product of these two vectors is zero.
Further $(6\hat{i} + 4\hat{j} - 10\hat{k})$ vectors is parallel to A. So, component of A along this vector is magnitude of A which is $5\sqrt{2}$ unit. The last vector i.e. $(-3\hat{i} - 4\hat{j} + 5\hat{k})$ is anti-parallel to A along this vector is negative of magnitude of A or $-5\sqrt{2}$ unit.

(103) ABC is an equilateral triangle. Length of each side is a and centroid is point O. Find $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} =$ .....



(A) 0 (B) 1

(C) 2 (D) 3

Solution:(Correct Answer:A)

$\overrightarrow{AB}, \overrightarrow{BC}$ and $\overrightarrow{CA}$ form a closed triangle in the same order.
$\therefore \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \overrightarrow{0}$

(104) A cricketer can throw a ball to a maximum horizontal distance of $100\,m$. With the same effort, he throws the ball vertically upwards. The maximum height attained by the ball is ..... m

(A) 100 (B) 80

(C) 60 **(D)** 50

Solution:(Correct Answer:D)

(d) Maximum range $= \frac{u^2}{g} = 100\,m$
Maximum height $= \frac{u^2}{2g} = \frac{100}{2} = 50\,m$

(105) A man moves in an open field such that after moving $10\,m$ on a straight line, he makes a sharp turn of $60°$ to his left. The total displacement just at the start of 8th turn is equal to .....m

(A) 12 (B) 15

(C) 17.32 (D) 14.14

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Solution:(Correct Answer:C)

(c)
When a man moves $10m$ before turning by $60°$, he makes a hexagon in six steps.
$\therefore$ In eight steps he will trace two additional stegs of hexagon from starting point
$\therefore$ Displacement is AB
$= \sqrt{10^2 + 10^2 + 2 \times 10 \times 10 \cos 60}$
$= \sqrt{100 + 100 + 200\left(\frac{1}{2}\right)}$
$= 10\sqrt{3} = 17.32\,m$

(106) Two forces, F_1 and F_2 are acting on a body. One force is double that of the other force and the resultant is equal to the greater force. Then the angle between the two forces is

(A) $\cos^{-1}(1/2)$ (B) $\cos^{-1}(-1/2)$

(C) $\cos^{-1}(-1/4)$ (D) $\cos^{-1}(1/4)$

Solution:(Correct Answer:C)

(c)Let $F1 = F$
and $F2 = 2F$
Resultant force $= F_{\text{net}} = 2F$
$F_{net} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2} \cos\theta$
$2F = \sqrt{F^2 + 2F^2 + 2 \times F \times (2F)} \cos\theta$
$4F^2 = 5F^2 + 4F^2 \cos\theta$
$4\cos\theta = -1$
$\therefore \cos\theta = \frac{-1}{4}$

(107) The displacement of a particle is proportional to the cube of time elapsed. How does the acceleration of the particle depends on time obtained

(A) $a \propto t^2$ (B) $a \propto 2t$

(C) $a \propto t^3$ **(D)** $a \propto t$

Solution:(Correct Answer:D)

(d) $x \propto t^3$
$\therefore x = Kt^3$
$\Rightarrow v = \frac{dx}{dt} = 3\,Kt^2$ and $\Rightarrow a = \frac{dv}{dt} = 6\,Kt$
i.e. $a \propto t$

(108) The magnitude of the X and Y component of $\vec{A}$ are 7 and 6 respectively. Also the magnitude of X and Y component of $\vec{A} + \vec{B}$ are 11 and 9 respectively. What is the magnitude of $\vec{B}$?

(A) 5 (B) 6

(C) 8 (D) 9

Solution:(Correct Answer:A)

Let $\vec{A} + \vec{B} = \vec{R}$
Given $A_x = 7$ and $A_y = 6$
$R_x = 11$ and $R_y = 9$
$\therefore B_x = R_x - A_x$
$= 11 - 7 = 4$
$B_y = R_y - A_y = 9 - 6 \& = 3$
$B = \sqrt{B_x^2 + B_y^2}$
$= \sqrt{4^2 + 3^2} = 5$

(109) A particle revolves round a circular path. The acceleration of the particle is

(A) Along the circumference of the circle

(B) Along the tangent

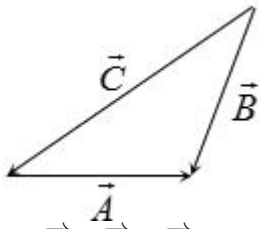
(C) Along the radius

(D) Zero

Solution:(Correct Answer:C)

Along the radius directed towards the centre.

(110) For the figure



(A) $\vec{A} + \vec{B} = \vec{C}$ (B) $\vec{B} + \vec{C} = \vec{A}$

(C) $\vec{C} + \vec{A} = \vec{B}$ (D) $\vec{A} + \vec{B} + \vec{C} = 0$

Solution:(Correct Answer:C)

(c)

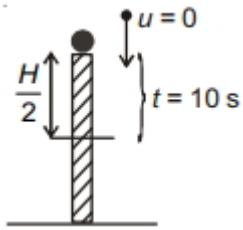
(111) A ball dropped from the top of tower falls first half height of tower in $10\,s$. The total time spend by ball in air is ..... s [Take $g = 10\,m/s^2$]

(A) 14.14 (B) 15.25

(C) 12.36 (D) 17.36

Solution:(Correct Answer:A)

(a)
$\frac{-H}{2} = ut - \frac{1}{2}g \times 10^2$
$\Rightarrow H = g \times 10^2$
$\Rightarrow -H = -\frac{1}{2}gt^2$ (Full joumey)
$g \times 10^2 = \frac{1}{2}gt^2$
$\Rightarrow t^2 = 200$
$\Rightarrow t = 10\sqrt{2}s$
$\Rightarrow t = 10 \times 1.414s$
$= 14.14s = t$



(112) A ball of mass m is thrown vertically upwards.Assume the force of air resistance has magnitude proportional to the velocity, and direction opposite to the velocity's. At the highest point, the ball's acceleration is

(A) 0 (B) less than g

(C) g (D) greater than g

Solution:(Correct Answer:C)

(113) The value of a unit vector in the direction of vector $A = 5\hat{i} - 12\hat{j}$, is

(A) $\hat{i}$ (B) $\hat{j}$

(C) $(\hat{i} + \hat{j})/13$ **(D)** $(5\hat{i} - 12\hat{j})/13$

Solution:(Correct Answer:D)

(d) $\vec{A} = 5\hat{i} + 12\hat{j}, |\vec{A}| = \sqrt{5^2 + (-12)^2} = \sqrt{25 + 144} = 13$
Unit vector $\hat{A} = \frac{\vec{A}}{|\vec{A}|} = \frac{5\hat{i} - 12\hat{j}}{13}$

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(114) A particle initially at rest is subjected to two forces. One is constant, the other is a retarding force proportional to the particle velocity. In the subsequent motion of the particle

(A) the acceleration will increase from zero to a constant value

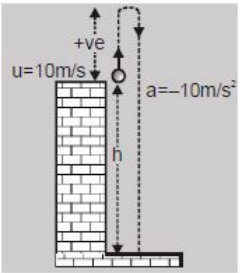
(B) the acceleration will decrease from its initial value to zero

(C) the velocity will increase from zero to a constant value.

(D) Both (B) and (C)

Solution:(Correct Answer:D)

(115) A ball is thrown upwards from the top of a tower $40\,m$ high with a velocity of $10\,m/s$, find the time when it strikes the ground .....sec : $(g = 10\,m/s^2)$



(A) 8 (B) 1

(C) 4 (D) 6

Solution:(Correct Answer:C)

$-40 = +10t - \frac{1}{2}(10)t^2$
$-4 = t - \frac{1}{2}t^2$
$t^2 - 2t - 8 = 0 \Rightarrow (t-4)(t+2) = 0$
$\therefore t = 4s$

(116) A projectile is thrown upward with a velocity v_0 at an angle α to the horizontal. The change in velocity of the projectile when it strikes the same horizontal plane is

(A) $v_0 \sin\alpha$ vertically downwards

(B) $2v_0 \sin\alpha$ vertically downwards

(C) $2v_0 \sin\alpha$ vertically upwards

(D) zero

Solution:(Correct Answer:B)

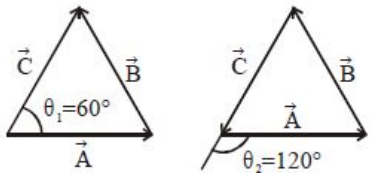
(b)
$\Delta a = a\Delta t (as$ a $=$ constant $)$
$= (-g\hat{j})\left(\frac{2v_0 \sin\alpha}{g}\right) = (-2v_0\alpha)\,\hat{j}$
i.e. change in velocity is $2v_0 \sin\alpha$, vertically downwards.

(117) Given that; $A = B = C$. If $\vec{A} + \vec{B} = \vec{C}$, then the angle between $\vec{A}$ and $\vec{C}$ is θ_1. If $\vec{A} + \vec{B} + \vec{C} = 0$, then the angle between $\vec{A}$ and $\vec{C}$ is θ_2. What is the relation between θ_1 and θ_2 ?

(A) $\theta_1 = \theta_2$ **(B)** $\theta_1 = \theta_2/2$

(C) $\theta_1 = 2\theta_2$ (D) None of these

Solution:(Correct Answer:B)



(118) Two bodies of masses $1kg$ and $5kg$ are dropped gently from the top of a tower. At a point $20cm$ from the ground, both the bodies will have the same

(A) Momentum (B) Kinetic energy

(C) Velocity (D) Total energy

Solution:(Correct Answer:C)

(c)Velocity of fall is independent of the mass of the falling body.

(119) A particle is rotating in a circle of radius $1\,m$ with constant speed $4\,m/s$. In time $1\,s$, match the following (in SI units) columns.

Colum I	Colum II
(A) Displacement	(p) $8\sin 2$
(B) Distance	(q) 4
(C) Average velocity	(r) $2\sin 2$
(D) Average acceleration	(s) $4\sin 2$

(A) $(A \to r, B \to q, C \to r, D \to p)$

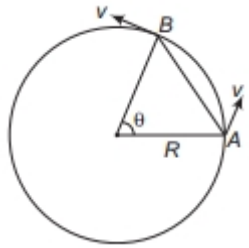
(B) $(A \to p, B \to q, C \to r, D \to p)$

(C) $(A \to r, B \to s, C \to r, D \to p)$

(D) $(A \to p, B \to q, C \to r, D \to s)$

Solution:(Correct Answer:A)

(a)
$\omega = \frac{v}{R} = \frac{4}{1} = 4\,rad/s$
$\theta = \omega t = 4\,rad$
Displacement $= AB = 2R\sin\frac{\theta}{2}$
$= 2\sin 2$ unit
Distance $= vt = 4$ unit
Average velocity $= \frac{\text{Displacement}}{\text{Time}} = 2\sin 2$ unit
And average acceleration $= \frac{|\Delta v|}{\Delta t}$
From subtraction of vectors, we can find that change in velocity from A and B is $2v\sin\frac{\theta}{2} = 8\sin 2$ unit
$\therefore$ Average acceleration $= \frac{8\sin 2}{1}$
$= 8\sin 2$ unit



(120) A particle is moving with constant speed $\sqrt{2}\,m/s$ on a circular path of radius $10\,cm$. Find the magnitude of average velocity when it has covered $\left(\frac{3}{4}\right)^{th}$ circular path

(A) $\frac{\pi}{3}\,m/s$ (B) $\frac{3}{2\pi}\,m/s$

(C) $\frac{3}{\pi}\,m/s$ **(D)** $\frac{4}{3\pi}\,m/s$

Solution:(Correct Answer:D)

$|<\overline{V}>| = \frac{R\sqrt{2}}{\left(\frac{3\pi R/2}{V}\right)} = \frac{2\sqrt{2}v}{3\pi}$
$= \frac{4}{3\pi}$m/s

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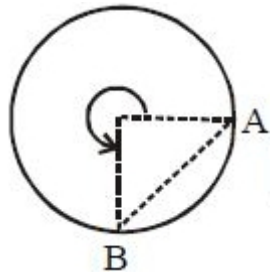


(121) Mark the wrong statement :-

(A) Nothing is in the state of absolute rest or stateof absolute motion.

(B) Magnitude of displacement is always equal to the distance travelled.

(C) Magnitude of displacement can never be greater than the distance travelled.

(D) Magnitude of instantaneous velocity is equal to the instantaneous speed.

Solution:(Correct Answer:B)

(122) A particle is moving in a circle with uniform speed v. In moving from a point to another diametrically opposite point

(A) The momentum changes by mv

(B) The momentum changes by $2mv$

(C) The kinetic energy changes by $(1/2)mv^2$

(D) The kinetic energy changes by mv^2

Solution:(Correct Answer:B)

(b) Momentum changes by $2\,mv$ but kinetic energy remains same.

(123) A string of length $2\ m$ can withstand a maximum tension of $16000\ N$. A particle of mass $2\ kg$ is tied to one end of the string and rotated in a horizontal circle. What is the maximum number of rotations per second the particle can make so that the string does not break? $[\pi^2 = g]$

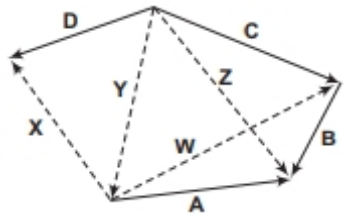
(A) 10 (B) 20

(C) 30 (D) 40

Solution:(Correct Answer:A)

$mr\omega_{\text{max}}^2 = T_{\text{max}}$
$\Rightarrow 2 \times 2 \times \omega_{\text{max}}^2 = 16000$
$\Rightarrow 4\pi^2 f_{\text{max}}^2 = 4000$
$\Rightarrow f_{\text{max}} = 10$ rotations per second

(124) In the diagram shown in figure



(A) $X = A + B - C + D$ (B) $Y = B + C - A$

(C) $Z = B + C$ **(D)** Only (b) and (c)

Solution:(Correct Answer:D)

(d)
Apply polygon law.

(125) The value of $\hat{i} \times (\hat{i} \times a) + \hat{j} \times (\hat{j} \times a) + \hat{k} \times (\hat{k} \times a)$ is

(A) a (B) $a \times \hat{k}$

(C) $-2\,a$ (D) $-a$

Solution:(Correct Answer:C)

(c)
Suppose $a = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$
Now, $(\hat{i} \times a) = a_2\hat{k} - a_3\hat{j}$
Now, $\hat{i} \times (\hat{i} \times a) = -a_2\hat{j} - a_3\hat{k}$
Similarly, $\hat{j} \times (\hat{j} \times a) = -a_1\hat{i} - a_3\hat{k}$
and $\hat{k} \times (\hat{j} \times a) = -a_1\hat{i} - a_2\hat{j}$
$\therefore \quad \hat{i}(\hat{i} \times a) + \hat{j}(\hat{j} \times a) + \hat{k}(\hat{k} \times a) = -2a$

(126) Two bodies of mass $10\,kg$ and $5\,kg$ moving in concentric orbits of radii R and r such that their periods are the same. Then the ratio between their centripetal acceleration is

(A) R/r (B) r/R

(C) R^2/r^2 (D) r^2/R^2

Solution:(Correct Answer:A)

(a)$\frac{a_R}{a_r} = \frac{\omega_R^2 \times R}{\omega_r^2 \times r} = \frac{T_r^2}{T_R^2} \times \frac{R}{r} = \frac{R}{r}\ [As\, T_r = T_R]$

(127) A car travels the first half of a distance between two places at a speed of $30\,km/hr$ and the second half of the distance at $50\,km/hr$. The average speed of the car for the whole journey is.....km/hr

(A) 42.5 (B) 40.0

(C) 37.5 (D) 35.0

Solution:(Correct Answer:C)

(c)Distance average speed$= \frac{2v_1v_2}{v_1+v_2} = \frac{2\times30\times50}{30+50}$
$= \frac{75}{2} = 37.5\ km/hr$

(128) A ball is dropped downwards. After 1 second another ball is dropped downwards from the same point. What is the distance between them after $3\,seconds$.....m

(A) 25 (B) 20

(C) 50 (D) 9.8

Solution:(Correct Answer:A)

(a) Distance between the balls = Distance travelled by first ball in 3 seconds -Distance travelled by second ball in 2 seconds $= \frac{1}{2}g\,(3)^2 - \frac{1}{2}g\,(2)^2 = 45 - 20 = 25\ m$

(129) In the $s - t$ equation $(s = 10 + 20t - 5t^2)$, match the following columns.

Colum I	Colum II
(A) Distance travelled in $3\,s$	(p) -20 units
(B) Displacement in $1\,s$	(q) 15 units
(C) Initial acceleration	(r) 25 units
(D) Velocity at $4\,s$	(s) -10 units

(A) $(A \to r, B \to p, C \to s, D \to p)$

(B) $(A \to p, B \to p, C \to s, D \to r)$

(C) $(A \to r, B \to s, C \to p, D \to p)$

(D) $(A \to s, B \to p, C \to r, D \to p)$

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Solution:(Correct Answer:A)

(a)
Comparing the given equation with general equation of displacement, $s = s_0 + ut + \frac{1}{2}at^2$, we get
$u = +20$ unit and $a = -10$ unit

(130) A car accelerates from rest at a constant rate of $2\ m/s^2$ for some time. Then it retards at a constant rate $4\ m/s^2$ and comes to rest. If it remains in motion for 3 second then displacement covered by it ....m

(A) 4 **(B)** 6

(C) 8 (D) 10

Solution:(Correct Answer:B)

$d = \frac{\alpha\beta t^2}{2(\alpha+\beta)} = 2 \times 4 \times 3$

(131) Two particles are at rest on a straight track and they are separated from each other by $100\,m$. They start their motion in the same direction, with the same acceleration $10\,m/s^{-2}$. Their relative displacement after $10\,s$ will be

(A) more than $100\,m$

(B) $100\,m$

(C) less than $100\,m$ but not zero

(D) zero

Solution:(Correct Answer:D)

$v_{A,B} = v_A - v_B$
Both A and B are at rest (initialy) So relative initial velocity will be zero
$a_{A,B} = a_A - a_B$
$= (10) - (10) = 0\ m/s^2$
relative acceleration will be zero.Relative Displacement will be Zero

(132) A disc rotating about its axis from rest, acquires the angular speed $100\,rev/s$ in 4 second. The angle rotated by it during these four seconds (in radian) is ..... π

(A) 100 (B) 200

(C) 300 **(D)** 400

Solution:(Correct Answer:D)

(d)
Initial angular speed, $\omega_0 = 0rad/s$
Final angular speed, $\omega = 100rps$
$\Rightarrow \omega = 100 \times 2\pi rad \Rightarrow \omega = 200\pi rad/s$
Time, $t = 4s$
From kinematics:
$\omega = \omega_0 + \alpha t$
$\Rightarrow 200\pi = 4\alpha$
$\Rightarrow \alpha = \frac{200\pi}{4} rad/s^2$
where, α is angular acceleration.
Also,
$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$
$\Rightarrow \theta = 0 + \frac{1}{2} \times \frac{200\pi}{4} \times 16$
$\Rightarrow \theta = 400\pi$

(133) The angle between two vectors given by $6\hat{i} + 6\hat{j} - 3\hat{k}$ and $7\hat{i} + 4\hat{j} + 4\hat{k}$ is

(A) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (B) $\cos^{-1}\left(\frac{5}{\sqrt{3}}\right)$

(C) $\sin^{-1}\left(\frac{2}{\sqrt{3}}\right)$ **(D)** $\sin^{-1}\left(\frac{\sqrt{5}}{3}\right)$

Solution:(Correct Answer:D)

(d) $\cos\theta = \frac{\vec{A}\vec{B}}{AB} = \frac{42+24-12}{\sqrt{36+36+9}\sqrt{49+16+16}} = \frac{56}{9\sqrt{71}}$
$\cos\theta = \frac{56}{9\sqrt{71}}$
$\therefore \sin\theta = \frac{\sqrt{5}}{3}$ or $\theta = \sin^{-1}\left(\frac{\sqrt{5}}{3}\right)$

(134) The average acceleration vector for a particle having a uniform circular motion is

(A) A constant vector of magnitude $\frac{v^2}{r}$

(B) A vector of magnitude $\frac{v^2}{r}$ directed normal to the plane of the given uniform circular motion

(C) Equal to the instantaneous acceleration vector at the start of the motion

(D) A null vector

Solution:(Correct Answer:D)

(d)In complete revolution change in velocity becomes zero so average acceleration will be zero.

(135) The position of a particle moving along the $y-$ axis is given as $y = 3t^2 - t^3$ where y is in $metre$ and t is in $second$. The time when the particle attains maximum positive position will be.....s

(A) 1.5 (B) 4

(C) 2 (D) 3

Solution:(Correct Answer:C)

$y = 3t^2 - t^3 \Rightarrow \frac{dy}{dt} = 6t - 3t^2 = 0$
at $t = 0; y = 0$
at $t = 2, y = 4$

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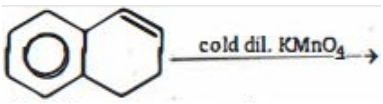
Chemistry - Section A (MCQ)

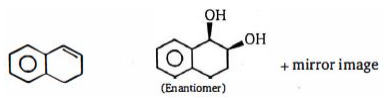
(136) Product of the reaction is



(A) Meso compound **(B)** Enantiomeric pair

(C) Diastereomers (D) Optically pure enantiomer

Solution:(Correct Answer:B)



(137) In a balanced equation
$H_2SO_4 + x\,HI \rightarrow H_2S + y\,I_2 + z\,H_2O$, the values of x, y, z are

(A) $x = 3, y = 5, z = 2$ (B) $x = 4, y = 8, z = 5$

(C) $x = 8, y = 4, z = 4$ (D) $x = 5, y = 3, z = 4$

Solution:(Correct Answer:C)

(c) The values of x, y, z are $8, 4, 4$ respectively hence the reaction is$H_2SO_4 + 8HI \rightarrow H_2S + 4I_2 + 4H_2O$

(138) Which of the following statement(s) is/are correct.
(A) The pH of $1 \times 10^{-8}M$ aqueous solution of HCl is 8
(B) The conjugate base of $H_2PO_4^{\ominus}$ is HPO_4^{2-}
(C) K_w increases with increase in temperature
(D) When a solution of weak monoprotic acid is

titrated against a strong base then at half neutralisation point, $pH = \frac{1}{2}pK_a$.

(A) A, D **(B)** B, C

(C) A, B, C (D) B, C, D

Solution:(Correct Answer:B)

$(B) \underset{\text{acid}}{H_2PO_4^{\ominus}} \underset{\text{C.B.}}{\rightarrow} HPO_4^{2-} + H^{\oplus}$

$(C)\ H_2O \rightleftharpoons H^+ + OH^{\ominus}$

Temp $\uparrow$ $K_w \uparrow$

(139) What will be the standard heat of reaction for the reaction for the reaction $C_2H_{6(g)} + 7/2O_{2(g)} \rightarrow 2CO_{2(g)} + 3H_2O_{(l)}$ if standard heat of combustion of $C_{(s)}$ and $H_{2(g)}$ are -393.5 and $-285\,kJ/mol$ respectively and standard heat of formation of $C_2H_{6(g)}$ is $-88.2\,kJ/mol$.....kJ/mol

(A) $+1553.8$ (B) -1730.2

(C) -1553.8 (D) -1456.6

Solution:(Correct Answer:C)

(140) When 2-bromobutane reacts with alcoholic KOH, the reaction is called

(A) Halogenation (B) Hydrogenation

(C) Chlorination **(D)** Dehydro-halogenation

Solution:(Correct Answer:D)

(d)

$CH_3 - \underset{\underset{Br}{|}}{CH} - CH_2 - CH_3 \xrightarrow{alc.KOH} \underset{\text{2-butene}}{CH_3CH = CHCH_3}$

The reaction is dehydrohalogenation.

(141) $(A) CH_3 - CH_2 - CH_2 - CO_2H$ $(B) CH_3 - \underset{\underset{CO_2H}{|}}{CH} - CH_3$

(A) and (B) are

(A) chain isomer (B) position isomer

(C) functional isomer (D) metamers

Solution:(Correct Answer:A)

(142) The lattice energy of $NaCl$ is $-780\,kJ\,mol^{-1}$. The enthalpies of hydration of $Na^+_{(g)}$ and $Cl^-_{(g)}$ ions are $-406\,kJ\,mol^{-1}$ and $-364\,kJ\,mol^{-1}$. The enthalpy of solution of $NaCl_{(s)}$ is.....$kJ\,mol^{-1}$

(A) 738 **(B)** 10

(C) -10 (D) -822

Solution:(Correct Answer:B)

$NaCl(s) \rightarrow Na^+(g) + Cl^-(g); \Delta H_1 = 780\,KJ$

$Na^+(g) + aq \rightarrow Na^+(aq) \Delta H_2 = -406\,KJ$

$Cl^-(g) + aq \rightarrow Cl^-(aq) \Delta H_3 = -364\,KJ$

$NaCl(s) + aq \rightarrow Na^+(aq) + Cl^-(aq)$

$\Delta H = \Delta H_1 + \Delta H_2 + \Delta H_3$

$= 780 - 406 - 364$

$= 10\,KJ\,mol^{-1}$

(143) Oxidation number of As atoms in H_3AsO_4 is

(A) -3 (B) $+4$

(C) $+6$ **(D)** $+5$

Solution:(Correct Answer:D)

(d) H_3AsO_4

$+3 + x - 2 \times 4 = 0;\ x = 8 - 3 = +5.$

(144) Octane number has 0 value for

(A) iso-octane (B) $n-$ hexane

(C) $n-$ heptane (D) iso-heptane

Solution:(Correct Answer:C)

n-heptane is defined as the zero point of the octane rating scale. $2,2,4-$Trimethylpentane has an octane rating of 100 , whereas n-heptane has an octane rating of 0. The octane number is based on an arbitrary scale which is a measure of its ability to resist knocking, as it burns on combustion.

(145) If degree of ionisation is 0.01 of decimolar solution of weak acid HA then pKa of acid is

(A) 5 (B) 7

(C) 2 (D) 3

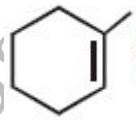
Solution:(Correct Answer:A)

$K_a = C\propto^2 = 0.1 \times (0.01)^2 = 10^{-5}$

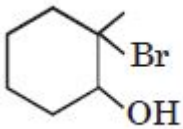
$pKa = 5$

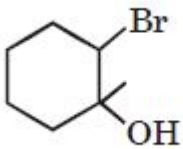
(146) What is major product of following reaction?

$\xrightarrow[H_2O]{Br_2}$

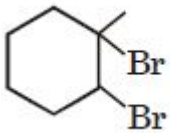


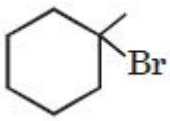
(A) **(B)**



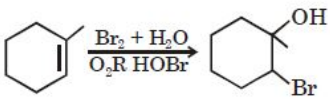


(C) (D)





Solution:(Correct Answer:B)



(147) $Me_2CH - \underset{\underset{OH}{|}}{CH} - Me \xrightarrow[350\,^oC]{Al_2O_3} (A) \xrightarrow[(ii)\ AgOH]{(i)\ HI} (B)$

Product (B) of above reaction

(A) $Me_2C(OH)CH_2Me$ (B) $Me_2CH - \underset{\underset{OH}{|}}{CH} - Me$

(C) $Me - \underset{\underset{OH}{|}}{CH} - CMe_3$

(D) $HO - CH_2 - (CH_2)Me$

Solution:(Correct Answer:A)

Vedclass

$$Me-\underset{\underset{OH}{|}}{\overset{\overset{Me}{|}}{CH}}-CH-Me \xrightarrow[\Delta,\,350\,^{o}C]{Al_2O_3}$$

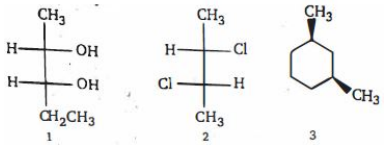
$$Me-\overset{\overset{Me}{|}}{C}=CH-Me \xrightarrow{HI}$$

$$Me-\underset{\underset{I}{|}}{\overset{\overset{Me}{|}}{C}}-CH_2-Me \xrightarrow[S_{N^1}(react.)]{AgOH}$$

$$Me-\underset{\underset{OH}{|}}{\overset{\overset{Me}{|}}{C}}-CH_2 \;+AgI$$

Al_2O_3 is dehydrating agent.

(148) Which of the following compounds are meso forms ?

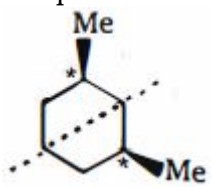


(A) 1 only **(B)** 3 only

(C) 1 and 2 (D) 2 and 3

Solution:(Correct Answer:B)

Meso $\rightarrow$ Compounds having chiral centres and have plane or centre of symmetry are known as meso compound.



(149) The $IUPAC$ name of the compound $CH_2=CH-CH(CH_3)_2$ is

(A) $1,1-$ dimethyl $-2-$ propene

(B) $2-$ vinyl propene

(C) $3-$ methyl but $-1-$ ene

(D) $1-$ isopropyl ethene

Solution:(Correct Answer:C)

$$\overset{1}{C}H_2=\overset{2}{C}H-\underset{\underset{CH_3}{|}}{\overset{3}{C}H}-\overset{4}{C}H_3$$

$3-$Methyl but$-1-$ene

(150) What is prefix of ester ?

(A) Oate (B) Haloformyl

(C) Carbalkoxy (D) Formyl

Solution:(Correct Answer:C)

(151) Which of the following is most reactive towards nucleophilic attack

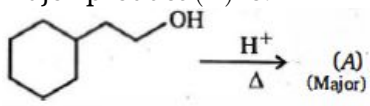
(A) $CH_3-\underset{\underset{O}{||}}{C}-OC_2H_5$ **(B)** $CH_3-\underset{\underset{O}{||}}{C}-Cl$

(C) $CH_3-\underset{\underset{OO}{||||}}{C}-O-C-CH_3$ (D) $CH_3-\underset{\underset{O}{||}}{C}-NMe_2$

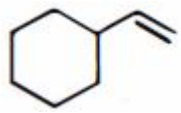
Solution:(Correct Answer:B)

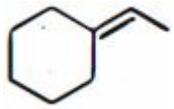
Rate of N_4 attack α electrophilicity of $>C=O$ group

(152) Major-product (A) is:

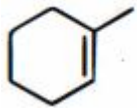


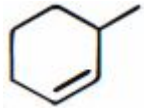
(A) **(B)**



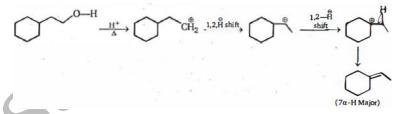


(C) (D)





Solution:(Correct Answer:B)

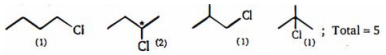


(153) The number of all types of isomers of chlorobutane is

(A) 2 (B) 4

(C) 6 **(D)** 5

Solution:(Correct Answer:D)



(154) If H^+ ion concentration of a solution is increased by 10 times its pH will be

(A) Increase by one (B) Remains unchanged

(C) Decrease by one (D) Increase by 10

Solution:(Correct Answer:C)

(c) When concentration of $[H^+]$increased then the value of pH is decreases.

$pH=\log\frac{1}{[H^+]}$

(155) In graphite $C-$atom is in ....state

(A) sp^3 (B) sp

(C) sp^2 (D) None of these

Solution:(Correct Answer:C)

Each carbon atom in graphiteis sp^2 hybridised and is bounded to three other carbon atoms.

(156) For the following gaseous reaction $H_2+I_2 \rightleftharpoons 2HI$, the equilibrium constant

(A) $K_p>K_c$ (B) $K_p<K_c$

(C) $K_p=K_c$ (D) $K_p=1/K_c$

Solution:(Correct Answer:C)

(c) If $\Delta n=0$ then $K_p=K_c$

Vedclass

(157) Find .....kJ/mol enthalpy of neutralisation of NH_4OH and HCN in aqueous solution if enthalpy of ionisation of NH_4OH and HCN are $7\ kJ/mol$ and $8\ kJ/mol$. also enthalpy of ionisation of H_2O is $57.3\ kJ/mole$

(A) -15 **(B)** -42.3

(C) $+1$ (D) 42.3

Solution:(Correct Answer:B)

(158) process $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)} + heat$

(A) $K_p = K_c$ (B) $K_p = K_c\,(RT)^{-1}$

(C) $K_p = K_c\,(RT)^{-2}$ (D) $K_p = K_c\,(RT)$

Solution:(Correct Answer:C)

The correct option is C

$K_P = K_C(RT)^{-2}$

For the reaction, $\Delta n = 2 - (1+3) = -2$, hence the correct relation would be $K_p = K_C(RT)^{-2}$

(159) Fischer Tropsch process is used for the manufacture of

(A) Synthetic petrol (B) Thermosetting plastics

(C) Ethanol (D) Benzene

Solution:(Correct Answer:A)

The Fisher-Tropscgh process is used in the manufacturing of Syntheticpetrol. The Fischer-Tropsch process is a collection of chemical reactions that converts a mixture of carbon monoxide and hydrogen into liquid hydrocarbons.... In the usual implementation, carbon monoxide and hydrogen, the feedstocks for FT, are produced from coal, natural gas, or biomass in a process known as gasification.

(160) The reaction $CH_3 - \underset{\underset{OH}{|}}{\overset{\overset{CH_3}{|}}{C}} - CH_3 \xrightarrow{H_2SO_4} CH_3 - \overset{\overset{CH_3}{|}}{C} = CH_2$ is the example of

(A) Sulphonation **(B)** Dehydration

(C) Alkylation (D) Decomposition

Solution:(Correct Answer:B)

(b) $$CH_3 - \underset{\underset{OH}{|}}{\overset{\overset{CH_3}{|}}{C}} - CH_3 \xrightarrow[Dehydration]{H_2SO_4}$$

$2-Methyl-2-hydroxypropane$

$$CH_3 - \overset{\overset{CH_3}{|}}{C} = CH_2 + H_2O$$

(161) Which one of the following series contain's electrophiles only ?

(A) $H_2O,\ SO_2,\ H_3O^{\oplus}$ (B) $NH_3,\ H_2O,\ AlCl_3$

(C) $AlCl_3,\ SO_3,\ \overset{\oplus}{N}O_2$ (D) $H_2O,\ C\overset{\oplus}{l},\ NH_3$

Solution:(Correct Answer:C)

Electophiles are electron deficient species. In other words, species having a tendency to accept electrons are called electrophiles. These may be neutral or positively charge, e.g., $AlCl_3, SO_3, NO_2$ etc.

While H_2O is having lone pair so it acts as lewis base not electron defecient. (Lewis acid)

(162) What is the weight of oxygen that is required for the complete combustion of $2.8\,kg$ of ethylene? ....kg

(A) 9.6 (B) 96

(C) 6.4 (D) 2.8

Solution:(Correct Answer:A)

$$\underset{(12\times2+4)}{C_2H_4} + \underset{96\ kg}{3O_2} \rightarrow 2CO_2 + 2H_2$$

$= 28kg$

Weight of oxygen required for the Complete combustion of $28kg$ ethylene $= 96kg$

$\therefore$ Weight of oxygen required for the combustion of $2.8kg$ ethylene.

$= \frac{96\times2.8}{28} = 9.6kg$

(163) Which of the following compounds which is an optically active compound

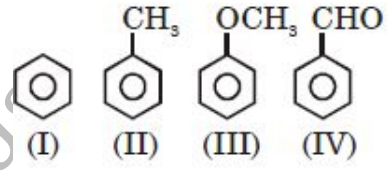
(A) $1-$butanol **(B)** $2-$butanol

(C) $3-$butanol (D) $4-$heptanol

Solution:(Correct Answer:B)

(b) $2-$butanol $(CH_3 - \underset{\underset{OH}{|}}{CH} - CH_2 - CH_3)$ is optically active because it has asymmetric carbon atom.

(164) Increasing order of electrophilic substitution for following compounds

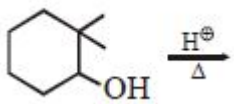


(A) $IV < I < II < III$ (B) $III < I < II < IV$

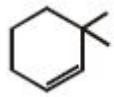
(C) $I < IV < III < II$ (D) $II < III < I < IV$

Solution:(Correct Answer:A)

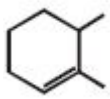
(165) Major product is



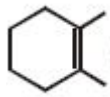
(A)



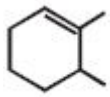
(B)



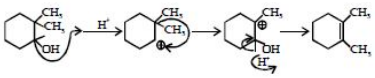
(C)



(D)



Solution:(Correct Answer:C)



(166) Consider the following reaction

$C_6H_6(l) + \frac{15}{2}\,O_2(g) \rightarrow 6\,CO_2(g) + 3H_2O\,(g)$

Signs of ΔH, ΔS and ΔG for the above reaction will be

(A) $+, -, +$ **(B)** $-, +, -$

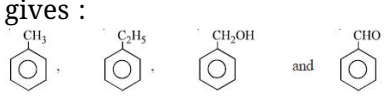
(C) $-, +, +$ (D) $+, +, -$

Vedclass

Solution:(Correct Answer:B)

$C_6H_6(l) + \frac{15}{2}O_2(g) \longrightarrow 6CO_2(g) + 3H_2O(g)$
Gaseous moles in reactant 7.5
Gareous moles in product $= 9$.
$\therefore \quad \Delta ng = 9 - 7.5 = +1.5$
$\therefore$ As Δng is tive $\therefore \Delta s$ its positive.
It is a combustion reaction (exothermic reaction)
$\therefore \Delta H$ is $-$ive.
$\Delta G_G = \underbrace{\Delta H}_{(-)} - T\underbrace{\Delta S}_{(+)}$
As T (temperature is positine)
$\therefore \quad \Delta G$ becomes negative.

(167) If the mixture of the following four aromatic compounds on oxidation by strong oxidising agent gives :



(A) Mixture of $C_6H_5CH_2OH + C_6H_5COOH$

(B) Mixture of $C_6H_5CHO + C_6H_5COOH$

(C) Only C_6H_5COOH

(D) None of the above

Solution:(Correct Answer:C)

(168) Which of the following compounds are not arranged in order of decreasing reactivity towards electrophilic substitution

(A) Fluoro benzene > chloro benzene > bromo benzene

(B) Phenol > $n-$propyl benzene > benzoic acid

(C) Chloro toluene > para-nitro toluene > $2-$chloro$-4-$nitro toluene

(D) Benzoic acid > phenol > $n-$propyl benzene

Solution:(Correct Answer:D)

(d) $-COOH$ group is a deactivating group, hence benzoic acid is less reactive towards electrophilic substitution than phenol.

(169) How many carbon atoms in the molecule $HCOO-(CHOH)_2-COOH$ are asymmetric

(A) 1 **(B)** 2

(C) 3 (D) None of these

Solution:(Correct Answer:B)

(b) $HCOO - \underset{(1)}{\overset{OH}{\overset{|}{\overset{*}{C}}H}} - \underset{(2)}{\overset{OH}{\overset{|}{\overset{*}{C}}H}} - COOH$

Two carbon atoms in the molecule are asymmetric.

(170) Which compound has the highest value of pK_a ?

(A) $Cl-CH_2-CH_2-COOH$

(B) CH_3-CH_2-COOH

(C) $CH_3-\underset{\underset{Cl}{|}}{CH}-COOH$

(D) $CH_3-\overset{\overset{Cl}{|}}{\underset{\underset{Cl}{|}}{C}}-COOH$

Solution:(Correct Answer:B)

(b) Acidic strength $\propto -I \propto \frac{1}{+I}$

(171) Hybridisation of carbon C_2 and C_4 respectively are $CH_3-CH=CH-CH_3$

(A) sp^2 and sp (B) sp and sp^2

(C) sp^3 and sp^2 **(D)** sp^2 and sp^3

Solution:(Correct Answer:D)

(172) Assign A, B, C, D from given type of reaction.

$NaNO_2 + FeSO_4 + H_2SO_4 \longrightarrow \underset{\text{(Ring complex)}}{[Fe(H_2O)_5NO]SO_4}$

(A) for disproportionation reaction.

(B) for comproportionation reaction.

(C) for either intermolecular redox reaction or displacement reaction.

(D) for either thermal combination redox reaction or thermal decomposition redox reaction.

Solution:(Correct Answer:C)

(173) In the process $BCl_3 + PH_3 \rightarrow BCl_3 : PH_3$ The Lewis acid is

(A) BCl_3 **(B)** PH_3

(C) Both (D) None

Solution:(Correct Answer:B)

(b)Because it accept electron pair from the PH_3

(174) For the equilibrium $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$, what is the temperature at which $\frac{K_p(atm)}{K_c(M)} = \frac{1}{3}$? .....K

(A) 0.027 (B) 0.36

(C) 36.54 **(D)** 4.06

Solution:(Correct Answer:D)

$\Delta n = 1$
$\frac{K_p}{K_c} = (RT)^{\Delta n} = 0.0821 \times T$
$\frac{1}{3} = 0.0821 \times T$
$\frac{1}{3 \times 0.0821} = T$
$4.06\,K = T$

(175) For the reaction $CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$, true condition is

(A) $K_p = K_c$ (B) $K_p > K_c$

(C) $K_p < K_c$ (D) $K_c = 0$ but $K_p \neq 0$

Solution:(Correct Answer:C)

(c) When $n_r > n_p$ then $K_p < K_c$
where $n_r =$ no. of moles of reactant
$n_p =$ no. of moles of product.

(176) H_2O_2 is used as

(A) An oxidant only

(B) A reductant only

(C) An acid only

(D) An oxidant, a reductant and an acid

Solution:(Correct Answer:D)

it's obvious.

Vedclass

(177) What is the correct $IUPAC$ name for

$$CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{H}{|}}{C}}-\ CH=CH-CH_2-\overset{\overset{O}{||}}{C}-OH$$

(A) $5-$ methyl $-3-$ hexenoic acid

(B) $5-$ carboxyl $-2-$ methylpentene

(C) $4-$ isopropyl $-3-$ butenoic acid

(D) None of above

Solution:(Correct Answer:A)
It's obvious.

(178) How many millilitres of $6.0\ M$ hydrochloric acid should be used to prepare $150\ ml$ of a solution which is $0.30\ M$ in hydrogen ion

(A) 3 **(B)** 7.5

(C) 9.3 (D) 30

Solution:(Correct Answer:B)
(b) $M_1 = 6.0\,M$ of HCl ; $V_1 =?$
$M_2 = 0.30\,M$ is H^+concentration in solution.
$V_2 = 150\,ml$ of solution.
$M_1V_1 = M_2V_2$; $6.0 \times V_1 = .30 \times 150$
$V_1 = \frac{.30\times150}{6} = 7.5\ ml.$

(179) Which one of the following reactions is not an example of redox reaction

(A) $Cl_2 + 2H_2O + SO_2 \rightarrow 4H^+ + SO^{4-} + 2Cl^-$

(B) $Cu^{++} + Zn \rightarrow Zn^{++} + Cu$

(C) $2H_2 + O_2 \rightarrow 2H_2O$

(D) $HCl + H_2O \rightarrow H_3O^+ + Cl^-$

Solution:(Correct Answer:D)
(d) In reaction $HCl + H_2O \rightarrow H_3O^+ + Cl^-$, only reduction has taken place not oxidation.

(180) The equilibrium concentration of X, Y and YX_2 are 4, 2 and 2 moles respectively for the equilibrium $2X + Y \rightleftharpoons YX_2$. The value of K_c is

(A) 0.625 **(B)** 0.0625

(C) 6.25 (D) 0.00625

Solution:(Correct Answer:B)
(b) $K_c = \frac{[YX_2]}{[X]^2[Y]} = \frac{2}{4\times4\times2} = \frac{1}{16} = 0.0625.$

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